Extracting Models from Data: A Cost-Benefit Framework*

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Abstract

We develop a theoretical framework for analyzing how learning occurs directly from data in the absence of prior information. We interpret *learning* as the process through which decision makers identify a *model*, defined as the set of variables that matter for the outcome, and use those variables to form *behavioral rules*. Rule extraction reflects a trade-off between each rule's value and its complexity cost. The comparative statics show that extraction of the highest-value rule depends not only on its own value, but also on how clearly it outperforms competing alternatives. Differences in cognitive costs then make agents respond differently to changes in relative rule values, generating systematic and predictable disagreement even when they face the same data. Finally, the framework makes rule-specific costs identifiable from behavior, allowing us to determine which objective measures of a rule's complexity can be interpreted as arguments of the cost function. We test the framework in our experiment and in a reanalysis of data from Kendall and Oprea (2024), finding strong support for its predictions in both settings.

Keywords: Learning, Model Misspecification, Complexity Costs, Bounded Rationality

JEL: C90, C92, D83, D90

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I INTRODUCTION

Individuals often form representations of how the world works by learning directly from data. Doing so requires determining which variables matter for outcomes and how they should guide predictions and actions, with the goal of identifying relationships that remain stable across environments. However, cognitive limitations and context-specific correlations may lead agents to learn relationships that appear useful in the environment in which they are observed but fail to generalize. Such errors can distort individual decisions (Esponda & Pouzo, 2016; Spiegler, 2016), sustain disagreement even under common information (Mailath & Samuelson, 2020), and undermine policy effectiveness by shaping how agents respond to policy-generated outcomes (Orphanides & Williams, 2008; Ito, 2014). This motivates our central question: how do the statistical properties of a dataset and, more broadly, features of the decision problem interact with individuals' cognitive limitations to shape what they learn from data? In particular, when do they recover stable relationships, and when do they instead learn correlations specific to the data at hand?

These inference problems arise across economic settings, where decision makers may extract patterns that are simple and fit the data at hand well, yet fail to capture the stable relationship governing outcomes. In macroeconomics, the Phillips curve captured a simple relationship between inflation and unemployment that fit historical data well but proved unstable across policy regimes, highlighting that a relationship useful for policy analysis must account for how agents' expectations respond to policy (Lucas, 1976). In financial markets, favorable public information about firms' revenues predicted future returns when trading frictions slowed price adjustment, but the relationship weakened as technology accelerated the incorporation of information into prices, revealing that the apparent regularity reflected features of the market environment rather than a stable relationship (Schwert, 2003; Ke & Ramalingegowda, 2005; Ng et al., 2008). The lesson is not simply that more data improve inference: evidence from different datasets may alter the relative fit of competing explanations and thereby help identify the underlying stable relationship. Yet, despite the economic importance and persistence of such errors (Schwartzstein, 2014; Giacomini et al., 2020; Grass et al., 2025), little is known about why particular misspecifications arise in the first place or how features of the decision problem shape what agents learn from data.

In this paper, we develop a theoretical framework to study agents who learn directly from data, in the absence of prior information, by selecting a set of variables for predicting the outcome (a *model*) and using those variables to form predictions or guide actions (a *behavioral rule*). We test the framework's novel predictions in a new experiment and in a reanalysis of data from Kendall and Oprea (2024). Building on cost–benefit models of reasoning (Alaoui & Penta, 2022) and the stochastic-choice framework of Chambers et al. (2025), the framework characterizes how the values and complexity costs of competing rules jointly shape whether

agents recover a stable relationship that generalizes across environments or instead adopt rules that fit only the data at hand. This distinction between value and cost organizes our main results. Beyond the role of absolute accuracy, we show that agents are more likely to recover the rule that generalizes when it more clearly outperforms competing rules. Moreover, reanalyzing the data of Kendall and Oprea (2024) while accounting for value-driven substitution, we find that only structural properties of rules, such as the number of variables they use, can be interpreted as entering the cost function. By contrast, commonly used complexity measures such as entropy capture variation in the relative value of competing rules rather than in their cost. By linking the primitives of the decision problem to testable predictions for rule extraction, our framework also yields additional comparative statics showing how incentives, the number of available variables, and the statistical properties of the dataset interact with complexity costs to shape what agents learn from data.

To formalize how the environment and cognitive costs jointly shape rule extraction and generalization, we introduce two key ingredients. First, we distinguish the stable relationship governing outcomes from dataset-specific correlations. We define a *dataset* as a joint distribution p over a finite collection of variables $X = (X_i)_{i \in N}$ and outcomes Y. We assume that outcomes are governed in every dataset by a stable *data-generating process* (DGP) $m^* : X_{M^*} \to \Delta(Y)$ that depends only on a set of relevant variables $M^* \subseteq N$.[1] The interpretation is that each dataset is generated by the same DGP m^* together with a potentially different distribution p_X over the variables, which may reflect exogenous variation across environments or be shaped endogenously by agents' previous choices.[2] Changes in p_X alter correlations among variables, so that variables included in a misspecified model $M \neq M^*$ may proxy well for the relevant variables in some datasets but not in others.

Second, we formulate the agent's problem as choosing which observables to use, selecting a model $M \subseteq N$, and extracting a behavioral rule $\pi_M : X_M \to \Delta(A)$ mapping those observables into actions. We formalize rule extraction as a trade-off between value and cognitive cost, building on the stochastic-choice framework of Chambers et al. (2025), yielding an endogenous form of Ockham's razor: simpler rules are favored when the additional value of more complex alternatives does not justify their higher cost.[3] The stability of the DGP

[1] The setup has a natural *causal* interpretation: environments may change the distribution and correlations among the variables while leaving the mechanism linking the relevant variables to the outcome unchanged (Pearl, 2009; Peters et al., 2016).

[2] For example, a stable demand relationship may map price and consumer characteristics into purchase probabilities, while the distribution of prices and characteristics differs across markets. Behavior may also endogenously shape the data available for inference: long-run action frequencies enter the distributions from which agents learn in Spiegler (2016) and determine the feedback used to assess misspecified models in Esponda and Pouzo (2016).

[3] We adapt the stochastic-choice framework of Chambers et al. (2025) directly to the space of rules, which yields closed-form extraction probabilities and facilitates identification of rule-specific costs. Online Appendix OE shows that our results extend to a broader class of additive perturbed utility models (Fudenberg et al., 2015).

implies the existence of a rule $\pi^{\max}_{M^*}$ that attains the highest value across datasets: because the conditional distribution of outcomes is the same for each $x_{M^*} \in X_{M^*}$ across datasets, so is the set of optimal actions for every realization.[4] The value of misspecified rules relative to the highest-value rule depends instead on how informative their variables are about those in M^*, and therefore on how often they reproduce the optimal action. By contrast, a rule's cost is stable across datasets and depends on objective features of the rule, such as the number of variables it uses, but may differ across individuals because of heterogeneity in attention, working memory, and cognitive control (Schneider & Shiffrin, 1977; Engle, 2002; Burks et al., 2009; Loewenstein & Wojtowicz, 2025). This separation allows us to derive comparative statics for how changes in the value of competing rules affect extraction while holding cognitive costs fixed (Alaoui & Penta, 2016; Alaoui et al., 2020; Alaoui & Penta, 2022). Because these costs are typically unobserved, we focus on predictions that do not require specifying the cost function. Once extraction probabilities are observed, however, rule-specific costs can be identified from choice data.[5]

Our first result highlights two dimensions of value that shape extraction of the highest-value rule: its absolute value and its advantage over competing alternatives. The first is intuitive: holding costs and competitors' values fixed, a higher value makes the rule more likely to be extracted. This provides a theoretical foundation for recent experimental evidence showing that reducing noise in the data makes the value-maximizing rule easier to recover (Fréchette et al., 2024). Our result goes further: what matters is not only how well the highest-value rule performs, but *how clearly* it outperforms its alternatives. When its advantage over every competing rule increases, it is more likely to be extracted (Proposition 1). Intuitively, when alternatives are close in value, it is difficult to distinguish among them; when one clearly stands out, choice becomes easier (Ratcliff & McKoon, 2008; Krajbich et al., 2010; Alós-Ferrer & Garagnani, 2022). However, greater separation can come at a cost: lowering the value of competing rules facilitates recovery of the highest-value rule but increases the losses from misspecification (Proposition 2). Thus, the environment that most clearly distinguishes the highest-value rule from its alternatives need not maximize average performance in the observed dataset (Milli et al., 2017; Raviv et al., 2022).

The separation between value and cost is also important for isolating how adding irrelevant variables, and thereby expanding the set of competing rules, affects extraction of the value-maximizing rule. A central idea in the complexity literature is that expanding the set of variables increases the space of possible representations and the cognitive burden of identifying the relevant relationship (Aragones et al., 2005; Gabaix, 2014; Sanjurjo, 2023; Kendall & Oprea, 2024). Our framework separates the effect of adding irrelevant variables from the possibility that those variables are observed in a new dataset, which can also alter

[4] While the highest-value rule is the same, its value can differ because datasets place different weight on realizations with different attainable payoffs.

[5] See Online Appendix OD for additional results when the cost function is known.

the relative values of existing rules. This sharpens the usual complexity intuition: holding the values of existing rules fixed, adding irrelevant variables enlarges the rule space and reduces extraction of the highest-value rule (Proposition 3). If, however, the new dataset makes the highest-value rule more clearly distinguishable from all alternatives, this may offset the effect of the larger rule space.[6]

The framework also generates systematic heterogeneity in rule extraction through differences in cognitive costs across agents. Lower-cost agents extract the value-maximizing rule more often and, when they make mistakes, place more weight on higher-value alternatives than higher-cost agents, who are also more likely to randomize (Proposition 4 and Corollary 2).[7] Together, these predictions provide a joint account of why more cognitively sophisticated individuals make fewer and better mistakes and respond more systematically to payoff-relevant features of the decision problem (Benjamin et al., 2013; Dean & Ortoleva, 2019; Guan et al., 2025), and why agents exposed to the same evidence can nevertheless extract different rules and reach different conclusions (Andre et al., 2022; Fréchette et al., 2024).

To test our theoretical predictions, we implement an experiment that isolates the core elements of the framework. Participants study several datasets, each generated by a different hypothetical machine. Each observation records the on/off status of several colored lights and whether the machine produces a sound. Participants may inspect the dataset and take notes, knowing that they will later be asked to predict whether the machine produces a sound for given light configurations using only those notes.[8] The design maps directly into the framework: the machine determines the conditional distribution of the sound given the relevant lights and therefore represents the stable *DGP*. For each machine, there is a *highest-value deterministic prediction rule* whose extraction maximizes prediction accuracy.[9] Each participant completes five tasks built around distinct machines and their associated highest-value rules, ensuring that the results are not specific to any particular machine. For a given machine, we vary the correlations among the lights across participants, generating

[6] Empirical asset pricing provides a prominent example: the proliferation of firm characteristics greatly expanded the set of predictors for firm returns, while new datasets showed that many apparent predictive relationships did not survive, helping distinguish structural relationships from sample-specific ones (McLean & Pontiff, 2016; Hou et al., 2020).

[7] Because lower- and higher-cost agents respond differently to relative rule values, changes in the decision environment can affect rule extraction differently across cost types. Using the measure of behavioral heterogeneity in Apesteguia and Ballester (2026), Proposition 8 in Online Appendix OD characterizes how disagreement varies with the statistical properties of the dataset.

[8] The experiment was conducted on Prolific with 253 participants from the United Kingdom and was pre-registered in the AEA RCT Registry (AEARCTR-0018501). The design builds on the "Blicket machine" paradigm in experimental psychology (Gopnik et al., 2004; Weisberg et al., 2020) and closely follows the experimental interface of Fréchette et al. (2024).

[9] For example, one of the machines produces a sound with probability 0.9 when the red light is on and 0.1 when it is off, independently of the other lights. The highest-value deterministic rule therefore predicts a sound if and only if the red light is on, attaining an accuracy of 90%.

different *datasets* in which both the accuracy of its highest-value rule and its advantage over competing rules change, while the identity of that rule remains fixed.

To isolate the two main channels through which rule values matter—the value-maximizing rule's advantage over competing rules and its absolute accuracy—we implement three conditions. Participants encounter datasets in which the value-maximizing rule is either highly accurate and has a large advantage over the best competing rule (*High Accuracy–High Gap*, HAHG), highly accurate but has only a small advantage (*High Accuracy–Low Gap*, HALG), or less accurate while retaining a large advantage (*Low Accuracy–High Gap*, LAHG).[10] In the treatment with two variables, a red and a blue light, extraction of the highest-value rule nearly doubles from 27% in HALG to 52% in HAHG as its advantage over competing rules increases, while its absolute accuracy remains unchanged. Consistent with the role of absolute accuracy, moving from HAHG to LAHG lowers the higher-value rule's accuracy while preserving a large advantage over competing rules, and reduces extraction by more than 10 percentage points. Finally, because the two forces work in opposite directions, the HALG–LAHG comparison provides an exploratory empirical test of their trade-off: without further restrictions on the cost function, the theory does not rank the two conditions. In the data, the larger value gap dominates the reduction in accuracy, increasing extraction of the highest-value rule by more than 10 percentage points.

To test the theory's prediction about the effect of expanding the set of available variables, we add a third irrelevant variable (a green light) while leaving the DGP and the values of all pre-existing rules unchanged. This treatment serves two purposes. It first replicates the value-gap result, showing that it is not specific to the two-variable environment. It then isolates the effect of the larger rule space: adding the irrelevant variable significantly reduces extraction of the highest-value rule, from 51.8% to 34.9% in HAHG and from 27.3% to 13.8% in HALG. This isolates the ceteris paribus effect emphasized by the theory: holding rule values fixed, more irrelevant variables create more competing rules and make the highest-value rule harder to extract (Aragones et al., 2005). The comparison between three-light HAHG and two-light HALG shows, however, that adding variables need not reduce its extraction when the value structure also changes. Despite the larger rule space, extraction is significantly higher in three-light HAHG, showing that a sufficiently larger value gap can more than offset the effect of additional variables. How strongly agents respond to these features of the environment, however, depends on the cognitive costs they face.

We capture heterogeneity in cognitive costs following Agranov et al. (2025), combining the time subjects spend studying the data, which measures effort, with their confidence in

[10] Averaged across machines, the accuracy-gap pairs are approximately 89% and 17 percentage points, 88% and 3 percentage points, and 82% and 12 percentage points in the three conditions, respectively. Each participant encounters every accuracy-gap condition at least once, allowing us to test the main predictions using within-subject variation.

their predictions. Effort alone is ambiguous: low effort can indicate either rapid understanding or giving up, while high effort can lead either to a confident solution or to failure to resolve the task. Confidence allows us to distinguish between these cases. Lower-cost subjects not only extract the highest-value rule more often but also make better mistakes, generating an upward shift in the entire distribution of extracted rule values. Structural estimates of the costs confirm this interpretation. Importantly, heterogeneity in cognitive costs generates systematic disagreement among individuals facing the same data (Andre et al., 2022; Fréchette et al., 2024), but does not overturn the main comparative statics: all cost types respond in the same direction to the value gap, absolute accuracy, and the number of variables.

The laboratory evidence therefore supports the framework's main predictions: rule values shape extraction, while cognitive costs shape agents' sensitivity to those values.

We next examine whether the same structure organizes behavior beyond our experimental design, using data from Kendall and Oprea (2024). In their experiment, each DGP is a deterministic finite automaton mapping histories into outputs. Crucially, each DGP is observed through different histories across subjects (*datasets* in our framework), generating exogenous variation in how well lower-value alternative automata (*rules* in our framework) predict outputs while holding the underlying DGP fixed. Whereas Kendall and Oprea (2024) focus on whether subjects recover the DGP, we extend their analysis using structural estimation (Dal Bó & Fréchette, 2011; Dvorak, 2022) to identify which alternative automata subjects extract when they fail to do so. Consistent with the value-gap prediction, the share attributed to lower-value alternative automata rises from 32.8% when they are less competitive to 50.9% when they are relatively valuable. We also find that subjects with greater mathematical training extract the DGP more often, place more weight on higher-value alternative automata, and are less likely to randomize, consistent with mathematical training proxying for lower cognitive costs.

The reanalysis also illustrates why separating value from cost is essential for interpreting objective measures of automaton complexity as arguments of the cost function (Alaoui & Penta, 2016, 2022). While Kendall and Oprea (2024) relate the complexity of the DGP to its extraction rate, our framework shows that this relationship need not identify automaton-specific complexity costs, because extraction also depends on substitution toward competing automata with different values. Once we account for all automata and their values, the associations with dataset-based complexity measures (such as entropy), which are naturally correlated with the values of competing automata, weaken considerably relative to those in Kendall and Oprea (2024), whereas associations with structural properties of automata, such as the number of states, remain robust. This pattern is consistent with structural features of rules, rather than statistical properties of datasets, being the relevant sources of complexity costs (Denti et al., 2022; Pomatto et al., 2023).

Finally, because the environment shapes what agents learn, a policy that is optimal when agents extract the highest-value rule may cease to be optimal once substitution towards lower-value rules is taken into account (Molnár & Santoro, 2014; Moll, 2026). The framework highlights two distinct implications. First, when policy does not alter the dataset agents observe, policymakers can improve behavior by lowering the complexity cost of extracting rules, for example through formal training and education (Fong et al., 1986; Lehman & Nisbett, 1990), or by presenting information in ways that make payoff-relevant relationships more salient (Kling et al., 2012; Hanna et al., 2014). Second, when policy itself shapes the data from which agents learn, endogenous learning can reverse policy rankings and should therefore be incorporated into policy design. A policy that is second-best when agents behave optimally may outperform an otherwise superior policy if it generates data that induce less substitution toward lower-value rules.

II RELATED LITERATURE

Our work contributes to a growing experimental literature on how individuals infer patterns directly from observed data, without prior information about the underlying data-generating process. Most closely related are Fréchette et al. (2024), Kendall and Oprea (2024), Fréchette et al. (2025), and Payzan-LeNestour et al. (2025), who study the mental models individuals form from abstract data generated by DAGs and finite automata, and document systematic departures from optimal inference. Our contribution to this literature is to develop a theoretical framework that delivers new empirical predictions and rationalizes the main findings on how rule extraction varies with the statistical structure of the data and task complexity (Fréchette et al., 2024; Kendall & Oprea, 2024). It moves the analysis from documenting patterns in rule extraction to deriving and testing comparative statics for how the decision environment shapes rule extraction.

The cost–benefit trade-off at the core of our framework connects to work in which the costs of reasoning interact with its benefits in shaping how deeply agents reason and which behavioral rules they adopt (Alaoui et al., 2020; Alaoui & Penta, 2022; Alaoui et al., 2026). This identification issue is central to Alaoui and Penta (2016), who show that observed depth of reasoning reflects both cognitive ability and financial incentives, so the distribution of play cannot by itself identify cognitive costs. Related cost–benefit trade-offs arise in information-structure choice (Guan et al., 2025), models of costly attention and rational inattention (Schwartzstein, 2014; Matějka & McKay, 2015; Dean & Neligh, 2023; Loewenstein & Wojtowicz, 2025; Ba et al., 2026), repeated games and strategy adoption (Rubinstein, 1986; Guan & Oprea, 2024), and communication and mechanism design (Van Zandt, 2007). By taking complete behavioral rules as the objects of choice, our framework extends this trade-off to rule extraction, making substitution across rules, and hence the misspecification that

emerges, endogenous to the primitives of the framework.

Because this value–cost trade-off determines which misspecification emerges, our framework also speaks to a growing theoretical (Hong et al., 2007; Esponda & Pouzo, 2016; Spiegler, 2016; Fudenberg et al., 2017; Heidhues et al., 2018; Mailath & Samuelson, 2020; Bohren & Hauser, 2021; Fudenberg et al., 2021) and empirical (Hanna et al., 2014; Enke, 2020; Esponda et al., 2024; Grass et al., 2025; Fan, 2026) literature on learning and decision making under misspecified models. Most closely related is the literature studying the consequences of incorrect causal representations, which focuses on directed acyclic graphs (DAGs) and on fitting outcome distributions under misspecified representations of causal structure (Spiegler, 2016, 2017, 2020; Ellis & Thysen, 2021).[11] While this literature studies the implications of adopting misspecified models and their persistence, we develop a tractable framework that characterizes which misspecifications arise in the first place. In doing so, we depart from approaches in which agents incorporate new information by updating beliefs within a prespecified class of models, and instead study learning directly from data in the absence of prior information.

A complementary theoretical (Eliaz & Spiegler, 2020; Schwartzstein & Sunderam, 2021; Aina, 2025) and experimental (Kendall & Charles, 2022; Ambuehl & Thysen, 2024; Aina & Schneider, 2025; Barron & Fries, 2025; Fan & Fries, 2026) literature studies how individuals select among exogenously supplied, potentially misspecified narratives.[12] In this literature, individuals are typically presented with data together with a set of candidate narratives, that is, alternative DGPs that could have generated the data, and decide which one to adopt. Unlike this literature, decision makers in our framework must extract DGPs directly from the data rather than evaluate candidate narratives supplied to them, thereby endogenizing the prior distribution over competing DGPs. This matters for learning under misspecification, where both the set of DGPs considered and their prior weights can determine which misspecifications persist in the long run (Esponda & Pouzo, 2016; Fudenberg et al., 2021).

The distinction between a stable DGP and dataset-specific correlations also links our framework to work in statistics and machine learning on out-of-domain generalization and invariant causal prediction (Peters et al., 2016; Rojas-Carulla et al., 2018; Zhou et al., 2023), which studies whether relationships learned in one distribution remain valid across environments that preserve the causal relationship between a subset of predictors and outcomes. The closest contribution in economics is Andrews et al. (2025), who develop finite-sample forecast intervals to assess how well a given model or algorithm transfers across domains. While the statistical logic is closely related, we take a behavioral perspective and study which

[11] A related theoretical literature studies coarse or robust extrapolation from data (Al-Najjar & Pai, 2014; Samuelson & Steiner, 2025), competition among models using different sets of explanatory variables (Montiel Olea et al., 2022), and the computational difficulty of discovering parsimonious, well-fitting theories (Aragones et al., 2005).

[12] For a recent review, see Ambuehl et al. (2026).

rule boundedly rational agents extract from the data. Because extraction depends on the relative values and cognitive costs of competing rules, changes across environments affect behavior not only by making the stable relationship easier or harder to distinguish, but also by changing the attractiveness of the alternatives agents may substitute toward.

Finally, because extraction depends on cognitive costs, the same mechanism provides a source of disagreement and heterogeneous expectations, consistent with evidence documenting substantial dispersion in forecasts and in the subjective models and narratives underlying them (Mankiw et al., 2004; Andre et al., 2022; Augenblick et al., 2025; Bastianello et al., 2025; Andre et al., 2026). Closer to our mechanism, Brock and Hommes (1997) and Branch and Evans (2006) endogenize the selection of forecasting rules through their relative performance, while Novák et al. (2024) and Vaccari (2024) generate disagreement through costly information processing or endogenous attention. We instead hold the observed dataset fixed and show how heterogeneity in cognitive costs can lead agents with common priors and preferences to extract different predictive rules and form different expectations.

III THEORETICAL FRAMEWORK

III(i) THE ENVIRONMENT

We begin with a familiar statistical environment. Let N denote a finite set of variables. For each $i \in N$, let $X_i \subseteq \mathbb{R}$ denote the finite set of possible realizations of variable i, with $X \equiv \prod_{i \in N} X_i$. We refer to any subset of variables $M \in \mathcal{P}(N)$ as a *model*.[13] For each model M, let $X_M \equiv \prod_{i \in M} X_i$ denote the set of possible realizations of the variables in M, and write $x_M = (x_i)_{i \in M}$ for a generic realization. Outcomes take values in a finite set $Y \subseteq \mathbb{R}$, with $y \in Y$ denoting a generic outcome.

We then distinguish the underlying *data-generating process* (DGP) from the *datasets* it generates. We assume that there exists a *true model* $M^* \subseteq N$ containing the variables relevant for outcomes. Conditional on the realizations of these variables, the DGP determines the distribution of outcomes. With a slight abuse of notation, we write[14]

$$m^* : X_{M^*} \to \Delta(Y).$$

A *dataset* is a joint distribution $p \in \Delta(X \times Y)$ over the variables and the outcome, with marginal $p_X \in \Delta(X)$ over the variables. We assume that each dataset is generated by

[13] $\mathcal{P}(N)$ denotes the power set of N. The empty set corresponds to a model with no variables.

[14] Formally, let $m^* : X \to \Delta(Y)$ denote the DGP. We require m^* to depend only on $X_{M^*} = \prod_{i \in M^*} X_i$, with M^* inclusion-minimal among the subsets satisfying this property. Thus, for every $x_{M^*} \in X_{M^*}$ and $x_{-M^*}, x'_{-M^*} \in X_{-M^*} = \prod_{i \in N \setminus M^*} X_i$, $m^*(\cdot \mid x_{M^*}, x_{-M^*}) = m^*(\cdot \mid x_{M^*}, x'_{-M^*})$. The model M^* therefore identifies the equivalence classes $\{x_{M^*}\} \times X_{-M^*}$ that are relevant for prediction.

combining the same DGP m^* with a potentially different p_X, which may vary across environments. The set of datasets consistent with m^* is

$$\mathcal{P}_{m^*} := \{p \in \Delta(X \times Y) : p(x,y) = p_X(x)m^*(y \mid x_{M^*}) \text{ for all } (x,y) \in X \times Y\}.$$

The framework gives the DGP a natural *causal* interpretation: datasets may differ in the joint distribution of the variables, and hence in their correlation patterns, while leaving the outcome-generating mechanism unchanged.[15] Importantly, datasets in our framework differ not because of finite-sample variation, but because they may feature different correlation patterns among the variables. As a result, the conditional distribution $p(\cdot \mid x_M)$ associated with a model M may generate predictions that closely match those of the DGP in some datasets but not in others.

Having specified the statistical environment, we now introduce the agent's *decision problem*. The agent chooses an action a from an action space A. Payoffs are represented by a utility function $u : A \times Y \to \mathbb{R}$, which depends on the chosen action and the realized outcome. This formulation encompasses a broad class of problems, including *prediction problems*, in which $A = Y$ (e.g., Kendall & Oprea, 2024); *belief-formation problems*, in which $A = \Delta(Y)$ (e.g., Brier, 1950; Danz et al., 2022); and more general problems, in which A and Y are different spaces.[16] A *decision problem* is a tuple $\mathcal{D} = (N, (X_i)_{i \in N}, Y, p, M^*, m^*, A, u)$, where $p \in \mathcal{P}_{m^*}$.

III(ii) RULE-BASED BEHAVIOR

We take a decision-theoretic approach and consider an agent who, given a model M, extracts a behavioral rule $\pi_M : X_M \to \Delta(A)$ mapping each realization $x_M \in X_M$ to a distribution over actions. The value of a rule π_M in decision problem $\mathcal{D}$ can be written as

$$v^{\mathcal{D}}(\pi_M) = \sum_{x_M \in X_M} p_X(x_M) \underbrace{\sum_{y \in Y} p(y \mid x_M) \int_{a \in A} \pi_M(da \mid x_M) u(a,y)}_{v^{\mathcal{D}}(\pi_M \mid x_M)},$$

and depends on two objects: the frequency with which each realization occurs, $p_X(x_M)$, and the rule's expected utility conditional on that realization, $v^{\mathcal{D}}(\pi_M \mid x_M)$. While many rules

[15] Through the lens of directed acyclic graphs (DAGs) (Spiegler, 2016), the *true model* can be interpreted as the set of parents of the outcome, the DGP as the conditional distribution of the outcome given those parents, and the *dataset* as the joint distribution over all nodes of the DAG. Online Appendix OF formalizes this interpretation and characterizes the conditions under which stability of the DGP across datasets suffices to identify the full set of causal parents of the outcome.

[16] In investment problems, Y may represent an asset payoff and A a set of portfolio positions. In policy problems, Y may represent economic outcomes and A the set of available interventions. In medical applications, Y may represent patient outcomes and A the set of possible treatments.

may be based on a given model M, a natural benchmark restricts attention to rules that are optimal given the information available to the agent. Under this benchmark, for each model M and realization x_M, the agent can be viewed as forming the conditional belief $p(\cdot \mid x_M)$ implied by the dataset and choosing an action that maximizes expected utility. Therefore, agents may adopt a misspecified model $M \neq M^*$ while still extracting a rule that is optimal given that model and the underlying dataset $p \in \mathcal{P}_{m^*}$. We denote by $\pi_M^{\max} : X_M \to \Delta(A)$ any rule satisfying this conditional optimality requirement.[17] While this value-maximizing benchmark connects directly to information economics (e.g., Frankel & Kamenica, 2019), our framework is more general and allows agents to extract rules that are not optimal conditional on a given model M.[18]

Because the DGP is invariant across datasets, $p(y \mid x_{M^*}) = m^*(y \mid x_{M^*})$ for every $p \in \mathcal{P}_{m^*}$. Hence, a rule $\pi_{M^*}^{\max}$ based on the true model M^* is value-maximizing in every dataset. For simplicity of exposition, we assume that this rule is unique. The rule's value, however, may change across datasets when they place different weights $p_X(x_{M^*})$ on realizations with different conditional values $v^{\mathcal{D}}(\pi_{M^*}^{\max} \mid x_{M^*})$, even though each conditional value is fixed across datasets. Similarly, the value gap between $\pi_{M^*}^{\max}$ and any alternative rule π_M is dataset-dependent and can be written as

$$v^{\mathcal{D}}(\pi_{M^*}^{\max}) - v^{\mathcal{D}}(\pi_M) = \sum_{x \in X} p_X(x) \underbrace{\sum_{y \in Y} m^*(y \mid x_{M^*}) \int_A u(a,y)[\pi_{M^*}^{\max}(da \mid x_{M^*}) - \pi_M(da \mid x_M)]}_{\Delta^{\mathcal{D}}(\pi_M \mid x_M, x_{M^*})}.$$

The value gap depends on two components. The conditional value gap $\Delta^{\mathcal{D}}(\pi_M \mid x_M, x_{M^*})$ measures how much value is lost by following π_M at x_M rather than acting optimally given x_{M^*}. The joint frequency $p_X(x)$ determines how often each gap occurs. Thus, a rule based on a misspecified model can remain close in value to the optimal rule either because it prescribes actions that are themselves close in value to the optimal actions or because the realizations on which it performs poorly are sufficiently rare in the dataset.

III(iii) THE STOCHASTIC CHOICE FUNCTION

Define the set of rules in decision problem $\mathcal{D}$ by $\Pi(\mathcal{D}) = \bigcup_M \Pi_M(\mathcal{D})$, where $\Pi_M(\mathcal{D})$ denotes a finite set of rules based on model M. Whenever no ambiguity arises, we write Π instead of $\Pi(\mathcal{D})$ to simplify notation. We assume that $\pi_{M^*}^{\max} \in \Pi$.

[17] Formally, $\pi_M^{\max}$ satisfies $\text{supp}(\pi_M^{\max}(\cdot \mid x_M)) \subseteq \arg\max_{a \in A} \sum_{y \in Y} p(y \mid x_M) u(a,y)$ for every $x_M \in X_M$. We assume that the decision problem is well defined, so that the relevant maximization problem admits a solution for every conditional belief.

[18] Online Appendix OB discusses the connection to information economics. Online Appendix OC illustrates the resulting flexibility with alternative inference procedures. For example, agents may encode probabilities imprecisely or coarsen their representations, as suggested by work on cognitive imprecision and rounding (Ruud et al., 2014; Woodford, 2020; Khaw et al., 2022).

Let $\rho^{\mathcal{D}} \in \Delta(\Pi)$ denote a probability measure over rules such that $\rho^{\mathcal{D}}(\pi_M) \geq 0$ for all $\pi_M \in \Pi$ and $\sum_{\pi_M \in \Pi} \rho^{\mathcal{D}}(\pi_M) = 1$. We now introduce the maximization problem in which, following Chambers et al. (2025), the probability distribution $\rho^{\mathcal{D}}(\cdot)$ is determined by

$$\rho^{\mathcal{D}}(\cdot) = \underset{\rho(\cdot)\in\Delta(\Pi)}{\operatorname{argmax}} \sum_{\pi_M \in \Pi} \left\{ v^{\mathcal{D}}(\pi_M) \cdot \rho(\pi_M) - \frac{\kappa(\pi_M)}{2} \cdot \rho(\pi_M)^2 \right\}. \tag{1}$$

Equation (1) describes agents who seek to maximize expected utility but face rule-specific costs, $\kappa(\pi_M) \in (0, \infty)$.[19] A key restriction is that $\kappa(\pi_M)$ depends only on the rule and is invariant across decision problems, capturing properties of the rule that govern the cost of internally representing the decision problem (*representational complexity*) and of manipulating that representation to reach a decision (*computational complexity*) (Ba et al., 2026; Oprea, 2026).[20] For example, the number of variables on which a rule π_M conditions provides a natural rule-specific measure of complexity that may enter the cost function, reflecting the idea that complexity rises with the number of informational components that must be processed or retained in working memory (Sanjurjo, 2023; Kendall & Oprea, 2024). This separability between costs and values is central to the framework: changes in the decision problem can alter rule values while leaving rule-specific costs fixed, yielding clean comparative statics for rule extraction (Alaoui & Penta, 2022).[21]

Solving the maximization problem in equation (1) yields a probability distribution over rules. As shown in Appendix A(ii) and Chambers et al. (2025), the solution may assign zero probability to some rules, with only those whose value exceeds an endogenous threshold entering the consideration set $\Pi^{\mathcal{D}}$.[22] For any rule $\pi_M \in \Pi^{\mathcal{D}}$, its probability of extraction is

$$\rho^{\mathcal{D}}(\pi_M) = \underbrace{\frac{\frac{1}{\kappa(\pi_M)}}{\sum_{\pi'_{M'}\in\Pi^{\mathcal{D}}} \frac{1}{\kappa(\pi'_{M'})}}}_{\text{Base probability term}} + \underbrace{\frac{1}{\kappa(\pi_M)} \left[v^{\mathcal{D}}(\pi_M) - \frac{\sum_{\pi'_{M'}\in\Pi^{\mathcal{D}}} \frac{1}{\kappa(\pi'_{M'})} v^{\mathcal{D}}(\pi'_{M'})}{\sum_{\pi'_{M'}\in\Pi^{\mathcal{D}}} \frac{1}{\kappa(\pi'_{M'})}} \right]}_{\text{Comparative value term}}. \tag{2}$$

Rule extraction is shaped by two distinct forces. The first is a base probability term, which

[19] The quadratic specification is not essential for the main comparative statics. Online Appendix OE provides a more general specification within the additive perturbed utility framework of Fudenberg et al. (2015) and states the assumptions on the cost function under which the results continue to hold.

[20] Appendix A(i) provides an alternative interpretation in which the cost captures the mental operations required to shift probability mass away from random choice toward rules that exploit information in the data.

[21] This restriction could be relaxed by allowing costs to be defined over equivalence classes of decision problems. Our comparative statics would continue to apply within an equivalence class, whereas comparisons across classes would require additional assumptions on how costs vary.

[22] We use the term consideration set differently from the literature (e.g. Manzini and Mariotti, 2014), since here it emerges endogenously rather than being specified exogenously. See Caplin et al. (2019) for a notable exception. Importantly, identification of the cost function yields testable predictions about which rules receive zero probability.

reflects how easy it is to notice and recall rule π_M relative to alternative rules, and depends on the costs. The second is a comparative value term, which captures the difference in value between a rule π_M and a weighted average of the values of all rules in the consideration set, scaled by the inverse of the cost: less costly rules are easier to notice, and so their value benefits (or drawbacks) are more easily noticed. Viewed through these two components, rule costs can be interpreted within our boundedly rational framework as reflecting scarce cognitive resources (Loewenstein & Wojtowicz, 2025; Ba et al., 2026), including limitations in working memory (Rustichini et al., 2023) and cognitive control (Schneider & Shiffrin, 1977; Posner & Snyder, 2004). Because these cognitive constraints vary across individuals (Engle, 2002; Burks et al., 2009), agents may face different cost functions.

The framework thus admits two interpretations. As in standard stochastic-choice models (Fudenberg et al., 2015), it can describe the stochastic behavior of a representative individual, or equivalently the aggregate distribution of rules in a homogeneous population. Alternatively, it can describe the distribution of rules within a group of agents sharing the same cost function, which is the interpretation we adopt when analyzing individual heterogeneity.

IV THEORETICAL RESULTS

To characterize the behavioral implications of the framework, we compare decision problems $\mathcal{D}$ and $\mathcal{D}'$ with the same highest-value rule $\pi^{\max}_{M^*}$. This comparison encompasses both the case in which the two problems share the same DGP m^* and the case in which they have distinct DGPs that nevertheless induce the same highest-value rule.[23] We derive comparative statics for its extraction probability and for substitution toward lower-value alternatives, focusing on sufficient conditions that deliver unambiguous comparative statics without imposing additional structure on the cost function. These conditions generate the predictions tested in our experiment. In Appendix D, we show how the cost function can be recovered from observed behavior, while Online Appendix OD derives additional comparative statics that become testable once the cost function is identified. All proofs are in Appendix A.

Rule Values. We start by characterizing how changes in the relative values of competing rules affect the probability of extracting the highest-value rule $\pi^{\max}_{M^*}$.

Proposition 1. *Consider two decision problems $\mathcal{D}$ and $\mathcal{D}'$ with the same candidate rule set Π and the same value-maximizing rule $\pi^{\max}_{M^*}$. Suppose that the value advantage of $\pi^{\max}_{M^*}$ over every alternative rule is strictly larger under $\mathcal{D}'$ than under $\mathcal{D}$, that is,*

$$v^{\mathcal{D}'}\left(\pi^{\max}_{M^*}\right) - v^{\mathcal{D}'}(\pi_M) > v^{\mathcal{D}}\left(\pi^{\max}_{M^*}\right) - v^{\mathcal{D}}(\pi_M) \qquad \text{for every } \pi_M \in \Pi \setminus \left\{\pi^{\max}_{M^*}\right\}.$$

[23] For example, the DGP may become more or less noisy (e.g. Fréchette et al., 2024), changing the rule's accuracy while leaving the optimal prediction unchanged.

Then $\rho^{\mathcal{D}'}\left(\pi^{\max}_{M^*}\right) \geq \rho^{\mathcal{D}}\left(\pi^{\max}_{M^*}\right) > 0$. *The inequality is strict whenever* $\rho^{\mathcal{D}}\left(\pi^{\max}_{M^*}\right) < 1$.

Proposition 1 establishes two results. First, the highest-value rule $\pi^{\max}_{M^*}$ is always extracted with positive probability. Second, it characterizes substitution across rules: as the highest-value rule's advantage over competing rules widens, agents become more likely to extract it.[24] As a special case, Proposition 1 captures the intuitive comparative static that increasing the value of $\pi^{\max}_{M^*}$, while holding the values of competing rules fixed, raises its extraction probability. While this comparative static is intuitive and consistent with the empirical evidence in Fréchette et al. (2024), we show that the increase in extraction is driven by the widening value gap. This is important because relative and absolute value need not move together: the highest-value rule may become more clearly separated from its alternatives even as all rules become less valuable.[25]

A natural way to vary the value gap is through incentives, for example by scaling the utility function. If $\pi^{\max}_{M^*}$ strictly outperforms every alternative, scaling all rule values up by a common factor widens its advantage over competing rules, while scaling them down narrows it. Corollary 1 characterizes this comparative static.

Corollary 1. *Let* $\mathcal{D}'$ *differ from* $\mathcal{D}$ *only in that* $u^{\mathcal{D}'}(a, y) = \alpha u^{\mathcal{D}}(a, y)$ *for all* $(a, y) \in A \times Y$ *and for some* $\alpha > 0$. *Then, if* $\pi^{\max}_{M^*}$ *is the unique value-maximizing rule in* $\mathcal{D}$*:*

- *If* $\alpha > 1$, *then* $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \geq \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$, *with strict inequality if* $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$, *and* $\Pi^{\mathcal{D}'} \subseteq \Pi^{\mathcal{D}}$. *Moreover, as* $\alpha \to \infty$, $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \to 1$.
- *If* $\alpha \in (0, 1)$, *then* $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \leq \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$, *with strict inequality if* $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$, *and* $\Pi^{\mathcal{D}'} \supseteq \Pi^{\mathcal{D}}$. *Moreover, as* $\alpha \to 0$, $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \to \frac{\frac{1}{\kappa(\pi^{\max}_{M^*})}}{\sum_{\pi_M \in \Pi} \frac{1}{\kappa(\pi_M)}}$.

In addition to providing a way to widen or narrow the value gaps, Corollary 1 shows that as α increases, the consideration set contracts monotonically under set inclusion, converging to the singleton containing the uniquely value-maximizing rule. Conversely, as α approaches zero, value differences become less consequential relative to costs, and the consideration set expands, dispersing probability across lower-value rules.

In passing, we note that the result also provides a choice-based method for identifying the highest-value rule when utility is not observed by the analyst. If the analyst can scale up stakes uniformly, the unique highest-value rule is the one whose extraction probability

[24] This parallels evidence-accumulation models, in which choices become faster and more accurate as alternatives become more sharply separated, and experimental findings that errors and choice instability rise as utility differences narrow (Ratcliff & McKoon, 2008; Krajbich et al., 2010; Alós-Ferrer & Garagnani, 2021, 2022).

[25] More generally, Proposition 5 in Online Appendix OD characterizes how substitution across rules depends on the cost function when the sufficient conditions of Proposition 1 do not hold.

converges to one. This argument parallels revealed-preference approaches that use incentive variation to recover costly information acquisition from stochastic choice (Caplin & Dean, 2015).

Proposition 1 implies that the highest-value rule is more likely to be extracted whenever its value advantage over every alternative widens, even if its own value falls, provided that competing rules lose still more value. However, in some classes of decision problems, the value of a rule that prescribes choosing an action at random remains constant across datasets.[26]

Proposition 2. *Let $\mathcal{D}$ and $\mathcal{D}'$ have the same candidate rule set Π. Suppose that $v^{\mathcal{D}'}(\pi^R) = v^{\mathcal{D}}(\pi^R)$, $v^{\mathcal{D}'}(\pi_M) < v^{\mathcal{D}}(\pi_M)$ for every $\pi_M \in \Pi \setminus \{\pi^R\}$, and that $\rho^{\mathcal{D}}(\pi^R) \in (0,1)$. Then the extraction probability of the random rule strictly increases, that is,*

$$\rho^{\mathcal{D}'}(\pi^R) > \rho^{\mathcal{D}}(\pi^R).$$

The proposition shows that when rule values decline while the value of π^R remains unchanged, extraction shifts toward randomization. Thus, in some decision problems, a decline in absolute rule values may widen the gap between $\pi^{\max}_{M^*}$ and competing rules while at the same time making randomization more attractive. In this case, the sufficient condition of Proposition 1 is no longer satisfied, and whether the highest-value rule is extracted more often depends on which force dominates and, more generally, on the cost function (Proposition 5 in Online Appendix OD).

The analysis so far evaluates learning by the probability of recovering the highest-value rule $\pi^{\max}_{M^*}$. This criterion is especially natural when the objective is generalization across datasets, since $\pi^{\max}_{M^*}$ is guaranteed to generalize across decision problems that share the same DGP m^*. Yet a higher extraction probability need not imply lower expected losses: as alternative rules become less valuable, $\pi^{\max}_{M^*}$ is extracted more often, but failures to extract it become more consequential. Such an analysis requires considering the expected value loss from rule extraction, defined as $\mathcal{L}^{\mathcal{D}} \equiv \sum_{\pi_M \in \Pi} \rho^{\mathcal{D}}(\pi_M) \left[v^{\mathcal{D}}\left(\pi^{\max}_{M^*}\right) - v^{\mathcal{D}}(\pi_M) \right]$, which can be rewritten as

$$\mathcal{L}^{\mathcal{D}} = \underbrace{\left[1 - \rho^{\mathcal{D}}\left(\pi^{\max}_{M^*}\right)\right]}_{\text{probability of an extraction error}} \underbrace{\sum_{\pi_M \neq \pi^{\max}_{M^*}} \frac{\rho^{\mathcal{D}}(\pi_M)}{1 - \rho^{\mathcal{D}}\left(\pi^{\max}_{M^*}\right)} \left[v^{\mathcal{D}}\left(\pi^{\max}_{M^*}\right) - v^{\mathcal{D}}(\pi_M) \right]}_{\text{average value loss conditional on an error}}. \tag{3}$$

This expression separates the probability of failing to extract the highest-value rule from the average value loss conditional on such a failure. Moving from $\mathcal{D}$ to $\mathcal{D}'$, greater extraction of $\pi^{\max}_{M^*}$ lowers the first margin, while changes in relative rule values and substitution across lower-value alternatives determine the second. When these margins move in opposite

[26] This includes the decision problem used in our experiment and formalized in Online Appendix OB(iii).

directions, the effect on expected value loss is generally ambiguous without knowledge of the rule-specific costs $\kappa(\cdot)$, which govern substitution toward lower-value rules and, in turn, the severity of extraction errors.

Rule Space. We next study how expanding the set of available rules, for example by adding irrelevant variables, affects rule extraction.

Proposition 3. *Suppose that $\mathcal{D}'$ differs from $\mathcal{D}$ only in that $\Pi(\mathcal{D}) \subset \Pi(\mathcal{D}')$, with $v^{\mathcal{D}'}(\pi_M) = v^{\mathcal{D}}(\pi_M)$ for every $\pi_M \in \Pi(\mathcal{D})$. Then $\rho^{\mathcal{D}'}(\pi_M) \leq \rho^{\mathcal{D}}(\pi_M)$ for every $\pi_M \in \Pi^{\mathcal{D}}$. The inequality is strict for every $\pi_M \in \Pi^{\mathcal{D}}$ if at least one newly available rule enters $\Pi^{\mathcal{D}'}$.*

Proposition 3 provides a regularity result in the spirit of Block and Marschak (1960): if two decision problems differ only in that $\mathcal{D}'$ expands the rule set available in $\mathcal{D}$, for example by adding irrelevant variables, then the extraction probability of every pre-existing rule weakly decreases, including that of the highest-value rule. This comparison applies when the values of all rules already available in $\mathcal{D}$ are held fixed. If these values also change in $\mathcal{D}'$, for example because the distribution of variables changes with the new dataset, the value-maximizing rule may become more clearly distinguishable from its alternatives. This can offset or even reverse the rule-space expansion effect, with the overall impact depending on $\kappa(\cdot)$. Thus, irrelevant variables are not harmful per se: they reduce extraction probabilities when their addition is the only change to the decision problem.

Cost function. We now turn to the role of costs in rule extraction. Although the same comparative statics apply to changes in an individual's cost function, we focus on heterogeneity across individuals, the dimension we study experimentally. For an individual of type j, let $\kappa_j(\cdot)$ denote the cost function, $\Pi_j^{\mathcal{D}}$ the induced consideration set in decision problem $\mathcal{D}$, and $\rho_j^{\mathcal{D}}$ the solution to the stochastic-choice problem in equation (1). Define the cumulative distribution of extracted-rule values by

$$F_j^{\mathcal{D}}(z) := \sum_{\substack{\pi_M \in \Pi: \\ v^{\mathcal{D}}(\pi_M) \leq z}} \rho_j^{\mathcal{D}}(\pi_M), \qquad z \in \mathbb{R}.$$

We say that $\rho_j^{\mathcal{D}}$ first-order stochastically dominates $\rho_{j'}^{\mathcal{D}}$, denoted $\rho_j^{\mathcal{D}} \succ_{\text{FOSD}}^{v^{\mathcal{D}}} \rho_{j'}^{\mathcal{D}}$, if $F_j^{\mathcal{D}}(z) \leq F_{j'}^{\mathcal{D}}(z)$ for every $z \in \mathbb{R}$, with strict inequality whenever $0 < F_{j'}^{\mathcal{D}}(z) < 1$. The following proposition characterizes how differences in cost functions shape rule extraction.

Proposition 4. *Let $\kappa_1(\cdot) = \gamma\kappa_2(\cdot)$, with $\gamma > 1$. Assume that $\Pi_1^{\mathcal{D}}$ contains rules attaining at least two distinct values. The following hold:*

(i) The low-cost type has a weakly smaller consideration set, that is, $\Pi_2^{\mathcal{D}} \subseteq \Pi_1^{\mathcal{D}}$.

(ii) There exists a threshold $\bar{v}^{\mathcal{D}} \in \left(\min_{\pi_M \in \Pi_1^{\mathcal{D}}} v^{\mathcal{D}}(\pi_M), v^{\mathcal{D}}(\pi_{M^*}^{\max})\right)$ *such that, for every* $\pi_M \in \Pi_1^{\mathcal{D}}$,

$$\operatorname{sgn}\left(\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\right) = \operatorname{sgn}\left(v^{\mathcal{D}}(\pi_M) - \bar{v}^{\mathcal{D}}\right).$$

For every $\pi_M \notin \Pi_1^{\mathcal{D}}$, $\rho_2^{\mathcal{D}}(\pi_M) = \rho_1^{\mathcal{D}}(\pi_M) = 0$.

(iii) $\rho_2^{\mathcal{D}} \succ_{\text{FOSD}}^{v^{\mathcal{D}}} \rho_1^{\mathcal{D}}$.

Point (i) shows that high-cost individuals have a larger consideration set. Although this expansion may look like a form of "open-mindedness", the additional rules that enter are precisely those with lower values, as made explicit in point (ii). More generally, as costs rise, the extraction probability of every rule above a threshold declines, while lower-value alternatives become more likely to be extracted. Point (iii) complements this result by showing that the distribution of extracted-rule values for lower-cost agents first-order stochastically dominates that for higher-cost agents.[27] The following corollary specializes these implications to two rules of particular interest: the highest-value rule $\pi_{M^*}^{\max}$ and the random rule π^R.

Corollary 2. *Under the assumptions of Proposition 4, the following holds:*

(a) If $\arg\max_{\pi_M \in \Pi} v^{\mathcal{D}}(\pi_M) = \{\pi_{M^*}^{\max}\}$, *then* $\rho_2^{\mathcal{D}}(\pi_{M^*}^{\max}) > \rho_1^{\mathcal{D}}(\pi_{M^*}^{\max})$.

(b) If $v^{\mathcal{D}}(\pi^R) < \bar{v}^{\mathcal{D}}$ *then* $\rho_2^{\mathcal{D}}(\pi^R) \leq \rho_1^{\mathcal{D}}(\pi^R)$, *with strict inequality if* $\rho_1^{\mathcal{D}}(\pi^R) > 0$.

Lower-cost types extract the value-maximizing rule more often, whereas higher-cost types are more likely to abandon systematic prediction in favor of the random rule. Thus, agents who observe the same dataset and share the same preferences and incentives may nevertheless differ in their extraction probabilities. Moreover, disagreement is endogenous to the decision problem rather than a fixed property of types. Different types may make very different choices in one environment, yet converge in another as changes in relative rule values affect their extraction probabilities differently.[28]

Cost identification. In Appendix D, we show how rule-specific costs can be identified from rule values and estimated extraction probabilities. This allows for further comparative statics and provides a disciplined way to evaluate the argument of the cost function.

[27] This extends first-order stochastic-dominance results from additive perturbed utility models to our framework (Fudenberg et al., 2015; Esteban-Casanelles & Gonçalves, 2026). These results also speak to recent experimental evidence on cognitive sophistication and the structure of mistakes. In particular, they rationalize the finding of Guan et al. (2025) that more sophisticated subjects make fewer errors overall, but that their remaining errors respond more systematically to features of the decision problem.

[28] Proposition 8 in Online Appendix OD uses the measure of behavioral heterogeneity proposed by Apesteguia and Ballester (2026) to formalize this point and characterize disagreement as a function of the framework's primitives.

Since the probability of extracting a given rule depends on the values and costs of its competing alternatives, recovering costs from data allows us to identify which objective measures of rule complexity enter the cost function and to test whether more complex rules are extracted less often, holding value fixed. This is important because substitution across rules may otherwise be attributed to complexity when it is instead driven by variation in the attractiveness of lower-value alternatives. In Section VII, we apply this logic by reanalyzing the data from Kendall and Oprea (2024), examining how the complexity of automata relates to their extraction probabilities.

V EXPERIMENTAL DESIGN

V(i) THE EXPERIMENTAL TASK

Each participant completes five tasks, each consisting of three parts. In Part 1, participants learn about an abstract machine that maps each light configuration into a conditional probability of the sound. In Part 2, participants observe additional realizations of the light configuration and, for each, predict whether the machine will produce a sound. In Part 3, we elicit participants' subjective assessments of the complexity of Parts 1 and 2.

Part 1: Observing the dataset. Each dataset is generated by an abstract machine, described as a device equipped with multiple colored lights and capable of producing a sound. In the language of our theoretical framework, each machine corresponds to a DGP that generates multiple observations. Each observation records the on/off status of every light and whether the machine produced a sound. Thus, the same machine can generate multiple datasets. We consider machines with either two lights, red and blue, or three lights, with the addition of a green light. Participants observe approximately 30 independent observations, allowing the entire dataset to be displayed on a single screen. Participants are informed that additional unobserved lights may influence the sound. This framing prevents them from ruling out probabilistic rules on purely logical grounds, without requiring us to introduce terms such as "stochastic" or "deterministic." Figure 1 presents a screenshot from an experimental session involving a dataset generated by a machine with two lights.

At the beginning of Part 1, participants are informed that, in Part 2, they will be asked to predict whether the machine produces a sound under different configurations of the lights. They are also told that the dataset will not be available in Part 2, but that they may take notes that remain accessible throughout the prediction task. Beyond this, they receive no information, except that observations are independent across rows. Each machine is



Figure 1: Example of a dataset in the two-light experiment

displayed for up to three minutes, after which participants are automatically advanced.[29]

To ensure that our findings do not hinge on any particular machine, each participant completes five tasks, each based on a dataset generated by a different machine. Although we consider machines that generate the sound probabilistically, the payment scheme induces a unique deterministic highest-value prediction rule for each machine, denoted by $\pi^{\max}_{M^*}$ in the theoretical framework. The five prediction rules we consider are as follows.

ALONE	Predict sound if and only if a specific light is on, regardless of the other lights.
OR	Predict sound if and only if at least one of the red and blue lights is on.
INHIBIT	Predict sound if and only if the red light is on and the blue light is off.
EITHER	Predict sound if and only if exactly one of the red and blue lights is on.
JOINT	Predict sound if and only if the red and blue lights have the same status.

When a third, green light is present, the conditional distribution of the sound continues to depend only on the red and blue lights. The green light is therefore irrelevant to both the DGP and its associated value-maximizing prediction rule.

Part 2: Prediction phase After studying the dataset, participants predict the sound for each possible light configuration multiple times. With two lights, there are $2^2 = 4$ distinct configurations, each presented three times. With three lights, there are $2^3 = 8$ configurations, each presented twice. Repeated observations under identical configurations

[29] The time constraint was rarely binding, with participants reaching the limit in only 12% of the two-light datasets and 25% of the three-light datasets.

allow us to classify each participant in each task from the pattern of their predictions and to distinguish systematic rule use from randomization with a high degree of accuracy.

Predictions are elicited sequentially and without feedback, and because the Part 1 dataset is no longer visible, participants must rely on the structure they have learned and on any notes-to-self recorded in Part 1, which remain available throughout the prediction task. Figure 2 provides a screenshot from the experiment.

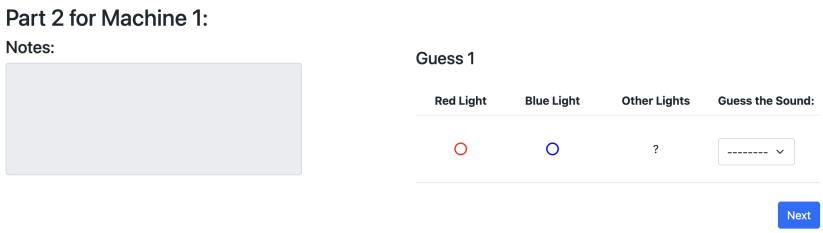


Figure 2: Example of a prediction in the two-light experiment

Part 3: Subjective assessment After completing each task, participants report (i) how many of their predictions they believe were correct, which provides a measure of self-assessed accuracy, and (ii) their confidence in that assessment. Together with the time spent on Part 1, these responses provide individual-level proxies of effort and confidence. We use these measures to construct an index of subjective task complexity adapted from Agranov et al. (2025) and to relate heterogeneity in perceived complexity to behavior, thereby testing the model's predictions regarding the cost function.

Payment. For each participant, one task is randomly selected at the end of the experiment. Participants earn £1.50 if a randomly drawn prediction from the task is correct, in addition to a fixed participation fee of £3.00 for the two-light experiment or £3.20 for the three-light experiment. The payment scheme can be represented as a simple decision problem within our theoretical framework, as formalized in Online Appendix OB.

V(ii) EXPERIMENTAL CONDITIONS

Each participant observes five datasets, one for each machine and its associated highest-value rule. A machine determines only the conditional distribution of the sound given the light configuration. We can therefore vary the distribution of light configurations to generate datasets with different correlation structures while holding the underlying DGP fixed. This variation leaves the highest-value rule unchanged while varying both its absolute accuracy and its advantage over lower-value alternatives. More specifically, our central experimental

manipulation focuses on two dimensions: the highest-value rule's absolute accuracy and the accuracy gap relative to the second-highest-value alternative.

For each machine, datasets vary along three dimensions: (i) the accuracy of the highest-value rule, (ii) the value gaps with the lower-value alternative rules, and (iii) the number of irrelevant variables. The design combines within- and between-subject variation. Each participant encounters each of the five machines in only one dataset, but the five tasks span different accuracy-gap conditions, allowing within-participant comparisons across conditions at the aggregate level. By contrast, comparisons across accuracy-gap conditions for a fixed value-maximizing rule are between participants. The number of lights is also assigned between subjects, with each participant completing all five tasks using either two-light or three-light machines. Comparisons that vary the number of irrelevant variables are therefore between participants. The accuracy-gap conditions, designed to test the comparative statics derived in Section IV, are as follows.

In the two-light experiment, for each value-maximizing rule, we vary the dataset to change both its absolute accuracy and its advantage over competing rules. In the *High Accuracy–Low Gap (HALG)* accuracy-gap condition, the DGP is highly accurate but only slightly outperforms the best alternative lower-value rule. The *High Accuracy–High Gap (HAHG)* and *Low Accuracy–High Gap (LAHG)* conditions feature substantially larger gaps: DGP accuracy remains high under HAHG and is lower under LAHG. In particular, the DGP is more accurate under HALG than under LAHG, but its advantage over competing rules is considerably smaller. This comparison allows us to distinguish the role of relative rule values from that of the DGP's absolute accuracy.

The three-light experiment implements the HALG and HAHG conditions. For each matched dataset, the joint distribution of the two original lights and the outcome is preserved, while a third, irrelevant light is added. Thus, the DGP and all pre-existing two-light rules retain the same values, but the additional variable expands the set of rules that subjects may extract. This comparison isolates the effect of expanding the variable space and provides a direct test of Proposition 3.

Each participant completes at least one task under every accuracy-gap condition in the experiment to which they are assigned. Accordingly, while comparisons for a fixed dataset are necessarily between subjects, the aggregate predictions of the theoretical framework can be tested within subjects by pooling observations across all five tasks. Moreover, we construct the experimental datasets so that the highest-value rule remains uniquely most accurate for every dataset.[30] The exact accuracies of the value-maximizing rule and the highest-value

[30] The only exception arises in the three-light HALG condition: within the unrestricted rule set, some non-DGP rules attain the same in-sample accuracy as the value-maximizing rule. These rules differ from the value-maximizing rule only for light configurations that are absent from the observed dataset or for which sound and no sound occur equally often. The highest-value rule remains uniquely most accurate among rules that condition on at most two lights.

alternative are reported in Table 1.[31]

	Two lights			Three lights	
	HALG	LAHG	HAHG	HALG	HAHG
Highest-value rule (%)	87.99	81.65	89.49	87.99	89.49
Second highest-value rule (%) ($\leq$ 2 lights / all rules)	84.65	70.03	72.97	84.65 / 87.99	73.60 / 82.88
Highest-value rule advantage (p.p.) ($\leq$ 2 lights / all rules)	3.34	11.62	16.51	3.34 / 0	15.89 / 6.61

Notes: The statistics are averages across the datasets generated by the five machines in each of the available accuracy-gap conditions. In the three-light HALG accuracy-gap condition the unrestricted rule set contains rules that attain the same in-sample value as the highest-value one. These rules differ from the highest-value one only at light configurations that are absent from the observed realizations or for which sound and no sound are equally frequent.

Table 1: Value structure by number of lights and accuracy-gap condition

We conducted three sessions of the two-light experiment with 152 participants and two sessions of the three-light experiment with 101 participants, for a total sample of 253. Table B1 in Appendix B summarizes the treatments to which participants were randomly assigned and the value configuration associated with each machine.

V(iii) IMPLEMENTATION

The experiment was implemented in oTree (Chen et al., 2016). Participants were recruited on Prolific from the United Kingdom; they were required to be native English speakers, aged 18–50, with at least a high-school diploma, and to have an approval rate of at least 99% on Prolific. After reading detailed instructions and passing a set of comprehension and AI checks (Çelebi et al., 2026), participants proceeded through the five tasks.[32] The study received ethical approval from the Committee on Institutional Review and Ethics at Universitat Pompeu Fabra (CIREP approval No. 468). Moreover, the experiment was pre-registered before data collection.[33]

[31] The broader distributional evidence in Online Appendix Table OA1 and Online Appendix Figure OA1 shows that the result is not driven by a single high-valued alternative. Specifically, HALG generates greater dispersion and a heavier upper tail. In the two-light design, the 10th–90th percentile range is 61.29 percentage points under HALG, compared with 37.86 under LAHG and 42.84 under HAHG. In the unrestricted three-light design, the interdecile range is 53.33 percentage points under HALG and 38.75 under HAHG.

[32] To deter AI-assisted participation, we presented an animated GIF in which a four-digit number scrolled vertically. Participants were required to identify and enter the number correctly. We also recorded response times and flagged implausibly fast submissions as potentially automated. Ex post, 98% of participants passed this check.

[33] The study was pre-registered in the AEA RCT registry, with pre-registration number: AEARCTR-0018501, available here.

V(iv) UNDERSTANDING THE DESIGN CHOICES

We designed the experiment with several objectives in mind.

First, following Fréchette et al. (2024), we adopt a deliberately context-free design in which datasets are presented as generated by abstract machines. Although this abstraction omits many features that shape learning in richer environments, it allows us to test the theoretical framework and study how participants extract rules from regularities in observed data without contextual confounds.

Second, the instructions are carefully formulated to avoid presenting participants with a predefined set of candidate models or a prior over such models. The design therefore places no ex ante restrictions on the representations participants may consider. This is deliberate, since our objective is to study how participants condition their predictions on observable variables, thereby revealing the statistical relationships they infer directly from the data. To avoid suggesting that any rule exists, the instructions also state that the lights and the sound may or may not be related.

Third, to keep the learning task manageable for participants, we restrict each dataset to approximately 30 observations. Moreover, the datasets are constructed to closely reflect the theoretical properties underlying our predictions. Drawing the observations independently for each participant would generate substantial cross-participant variation in the realized data. While such variation may be informative, it would impede the uniform satisfaction of the sufficient conditions required for our theoretical results.

Fourth, the set of machines and associated rules is chosen to avoid extremes of difficulty. We therefore restrict attention to binary variables and outcomes, and to machines with at most two relevant variables. For the same reason, we limit each participant to five machines to keep the overall cognitive burden manageable.

Fifth, we randomize both the order of observations within each dataset and the order in which datasets are presented across participants. This limits the scope for order-dependent effects, including contrast-based salience (Bordalo et al., 2013, 2022), to confound the relationship between rule complexity and behavior.

Finally, no feedback is provided between predictions in Part 2. If subjects received outcome feedback after each prediction, this information could spill across tasks: learning which rule is correct in one task may affect which rules subjects entertain in subsequent ones, contaminating the within-subject comparisons that underlie our main analysis.

VI RESULTS

This section reports the experimental results. We begin by using subjects' predictions to test whether they extracted the highest-value rule from each dataset.[34] For each subject–dataset pair, we classify the subject as having *extracted* the highest-value rule if an exact binomial test rejects, at the 1% level, the null hypothesis that agreement with the rule results from independent, uniformly random choices.[35] Among subjects who do not meet this criterion, we classify a subject as *randomizing* on that dataset if a separate exact test fails to reject random choice at the 5% level.[36] Figure 3 plots, for each experimental condition, the shares of observations consistent with extraction of the highest-value rule and with randomization, pooling across subjects and the five machines.[37]

Panel 3a reports highest-value rule extraction in the two-light experiment. The theoretical framework highlights two forces shaping extraction: how clearly the highest-value rule outperforms its alternatives and how accurate it is in absolute terms. Holding the highest-value rule's accuracy essentially unchanged, widening its value gap (HALG vs. HAHG) almost doubles extraction, from 27.3 to 51.8 percent ($p < 0.01$).[38] Conversely, preserving a large value gap while lowering absolute accuracy (HAHG vs. LAHG) reduces extraction from 51.8 to 40.9 percent ($p < 0.01$). While these two comparisons are consistent with Proposition 1, when absolute accuracy falls as the value gap increases (HALG vs. LAHG), the theory does not rank the two conditions ex ante without knowledge of the cost function. In the data, highest-value rule extraction rises from 27.3 to 40.9 percent ($p < 0.01$), suggesting that the larger value gap more than offsets the decline in absolute accuracy. These findings refine the notion of noise in rule extraction: extraction depends not only on the value-maximizing rule's absolute accuracy (Fréchette et al., 2024), but also on how clearly it stands out from plausible alternatives (Ratcliff & McKoon, 2008).

Recovering the highest-value rule more often does not necessarily improve average in-sample prediction performance: widening its separation from competing rules (HALG vs. HAHG) makes the DGP-implied rule substantially easier to recover, but the extraction failures that remain become more consequential. Table B2 in Appendix B applies the decom-

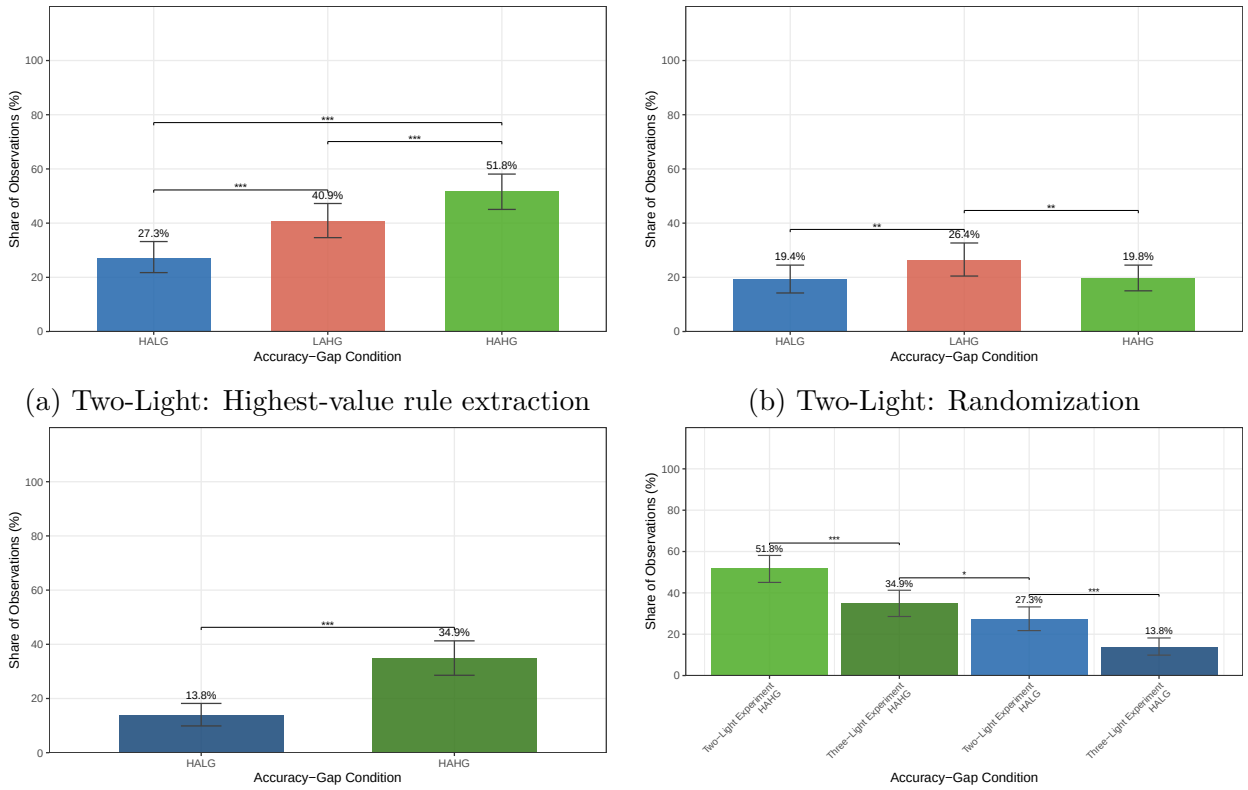
[34] See Online Appendix OA(ii) for the analysis of the written notes, which provide complementary evidence on the processes underlying subjects' decisions.

[35] Meeting the extraction criterion requires 11 correct predictions out of 12 in the two-light experiment and 14 out of 16 in the three-light experiment.

[36] We construct the exact null distribution of the test statistic under random choice, accounting for the search across alternative rules. Appendix D(i) provides details.

[37] The within-subject comparison is possible because each subject contributes at least one observation to each accuracy-gap condition. Appendix B reports the corresponding results separately for each machine.

[38] We assess pairwise differences using unadjusted, two-sided tests based on centered bootstrap distributions. Standard errors are computed from 5,000 subject-level bootstrap replications, resampling subjects with replacement within each experimental version and retaining all observations for each sampled subject. Appendix D(i) provides further details.



(a) Two-Light: Highest-value rule extraction

(b) Two-Light: Randomization

(c) Three-Light: Highest-value rule extraction

(d) Two- vs Three-Light: Highest-value rule extraction

Figure 3: Highest-value rule extraction and randomization across accuracy-gap conditions. Bars report the shares of subject–dataset observations classified as extracting the highest-value rule (exact binomial test) or as consistent with random choice (exact conditional test based on the best-fitting alternative) in the two- and three-light experiments. Error bars show 95% subject-level bootstrap confidence intervals.

position in equation (3) and makes this trade-off transparent. In the two-light experiment, fewer extraction failures are offset by their greater severity, leaving expected extraction loss similar across the two conditions, at 9.3 and 10.5 percentage points.

A second consequence of lower absolute accuracy is a shift toward randomization. When a larger value gap comes at the cost of lower absolute accuracy (HALG vs. LAHG), the highest-value rule is recovered more often, improving generalization, but alternative deterministic rules perform worse in the observed dataset, making randomization more attractive, as captured by Proposition 2. Panel 3b shows this margin directly: the share of observations consistent with random choice rises to 26.4 percent in LAHG, compared with 19.4 percent in HALG and 19.8 percent in HAHG ($p < 0.05$ for both comparisons).

We next ask whether the value-gap effect survives when the rule space expands. Panel 3c shows that it does: when the highest-value rule is more clearly separated from its alternatives (HALG vs. HAHG), extraction rises from 13.8 to 34.9 percent ($p < 0.01$). Consistent with

the framework, whose predictions do not depend on the number of variables, the value-gap effect therefore persists even when an additional irrelevant variable substantially expands the set of candidate rules.

Panel 3d then isolates the effect of expanding the rule space with an irrelevant variable. Holding fixed the joint distribution of the original two lights (Red and Blue) and the sound, we add an uninformative third light (Green), which creates additional rules without changing the values of the pre-existing ones. Consistent with Proposition 3, adding this irrelevant variable reduces highest-value rule extraction both when the rule is well separated from its alternatives (2L HAHG vs. 3L HAHG), from 51.8 to 34.9 percent ($p < 0.01$), and when the value gap is small (2L HALG vs. 3L HALG), from 27.3% to 13.8% ($p < 0.01$). The effect of a larger rule space, however, can be offset by greater separation in value. When the additional variable is paired with a larger value gap (2L HALG vs. 3L HAHG), extraction is higher despite the larger rule space, at 34.9 rather than 27.3 percent ($p < 0.10$). The theory does not rank this comparison ex ante because the two forces move in opposite directions: the additional variable expands the set of candidate rules, while the larger value gap makes the highest-value rule easier to distinguish from them. Thus, a larger variable space does not necessarily make rule extraction harder when it is accompanied by greater separation between the highest-value rule and its alternatives.

VI(i) RULE SUBSTITUTION AND SUBJECT HETEROGENEITY

The binomial tests are well suited to testing our main hypotheses, but they do not reveal which alternative rules subjects substitute toward when they fail to extract the highest-value rule, or how this substitution varies with heterogeneity in costs. To study both margins, we turn to a richer classification of behavior. Following Dal Bó and Fréchette (2011), we use maximum likelihood to estimate the population shares of candidate rules and use these estimates to construct posterior probabilities over rules for each subject–dataset pair.[39] We then proxy for heterogeneity in costs using a measure of subjective task difficulty adapted from Agranov et al. (2025). For each accuracy-gap condition, the measure combines effort, measured by time spent in the learning stage, with confidence in predictive performance.[40] Subjects who reach high confidence with little effort appear to find the task *Easy*; those who reach high confidence only after substantial effort find it *Doable*; those who remain uncertain

[39] The candidate set contains all deterministic mappings—16 with two lights and 256 with three—together with stochastic strategies that do not condition on the lights. We use model selection only to exclude empirically unsupported strategies. Estimation uses the `stratEst` package for *R* (Dvorak, 2022). Appendix D details the maximum-likelihood estimation and shows that it confirms the main findings from the binomial tests.

[40] Within each experiment and accuracy-gap condition, we classify effort and confidence relative to their corresponding means. Appendix D(iii) describes the procedure in detail. Figure B5 in Appendix B(ii) shows that these classifications are persistent across accuracy-gap conditions: subjects assigned to a given type in one condition are substantially more likely to receive the same classification in the others.

despite substantial effort find it *Hard*; and those who exert little effort and remain uncertain are classified as *Too Hard*, consistent with disengaging from a task perceived as too difficult to justify further effort.

Table 2 brings these two pieces together and reports the main evidence on substitution across rules and heterogeneity across cost types. Columns (1) and (4) use the posterior probabilities from the maximum-likelihood model to ask whether subjects substitute away from lower-value rules, and whether this sensitivity to value differs across cost types. Columns (2), (3), (5), and (6) use the exact-test classification underlying Figure 3 to examine how highest-value rule extraction and randomization vary across cost types and experimental conditions.

Experiment	Two-Light Experiment			Three-Light Experiment		
Outcome	All Rules	True Extracted	Randomizing	All Rules	True Extracted	Randomizing
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	29.8***	22.3***	22.1***	11.5***	5.47*	33.5***
	(1.33)	(3.53)	(3.62)	(0.735)	(3.12)	(6.48)
Condition: LAHG	-0.266*	14.0***	6.47**			
	(0.158)	(3.83)	(3.13)			
Condition: HAHG	1.42***	23.9***	0.931	2.32***	20.6***	0.767
	(0.443)	(3.94)	(3.16)	(0.475)	(3.85)	(3.86)
Distance (p.p.)	-0.641***			-0.152***		
	(0.061)			(0.047)		
High effort, low confidence	4.47**	0.651	4.93	1.22	-0.266	2.87
	(1.85)	(5.31)	(5.38)	(1.15)	(4.35)	(8.01)
Low effort, high confidence	1.72	9.70	-12.3***	0.752	14.6**	-18.3**
	(2.30)	(5.92)	(4.18)	(1.49)	(6.38)	(8.14)
High effort, high confidence	5.61**	18.3***	-9.80**	3.79***	22.8***	-17.3**
	(2.34)	(6.06)	(4.62)	(1.14)	(4.99)	(7.13)
Distance × High effort, low confidence	-0.197**			-0.063		
	(0.081)			(0.066)		
Distance × Low effort, high confidence	-0.056			-0.047		
	(0.098)			(0.079)		
Distance × High effort, high confidence	-0.256**			-0.225***		
	(0.105)			(0.062)		
Subjects	152	152	152	101	101	101
Observations	4,761	760	760	5,095	505	505
R^2	0.126	0.061	0.032	0.019	0.114	0.048

Notes: Standard errors in parentheses are heteroskedasticity-robust and clustered by subject. The *All Rules* columns use posterior assignment probabilities, so each subject enters once for every candidate rule in each dataset; these probabilities sum to one within each subject–dataset pair. The other columns use assignments from exact statistical tests. The omitted subject type is low effort–low confidence, and HALG is the omitted condition in both experiments.
$^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 2: Rule Extraction by Accuracy and Subjective Task Difficulty

Columns (1) and (4) of Table 2 reveal two important patterns. First, the farther a rule's accuracy falls below that of the highest-value rule, the less likely subjects are to extract

it, with a significant relationship in both the two- and three-light experiments. Second, subjects facing lower costs respond significantly more strongly to differences in rule value in both experiments, consistent with a complementarity between value and cost: differences in value matter more for choice when costs are lower.[41]

The same result emerges from columns (2), (3), (5), and (6), which use the binary outcomes underlying Figure 3. Controlling for subject type leaves the condition effects essentially unchanged, showing that the value-gap effects are present across types. In particular, highest-value rule extraction remains significantly higher when the value gap is larger, while lower absolute accuracy increases randomization. At the same time, the regressions show systematic heterogeneity in how subjects respond to the task. Subjects who find the task *Easy* or *Doable* are less likely to randomize and more likely to extract the highest-value rule than those who find it *Too Hard*, whereas subjects who find the task *Hard* behave more similarly to the *Too Hard* group on these margins. Together with columns (1) and (4), these results support Proposition 4: lower costs increase extraction of the highest-value rule and, more generally, shift choice toward higher-value rules.

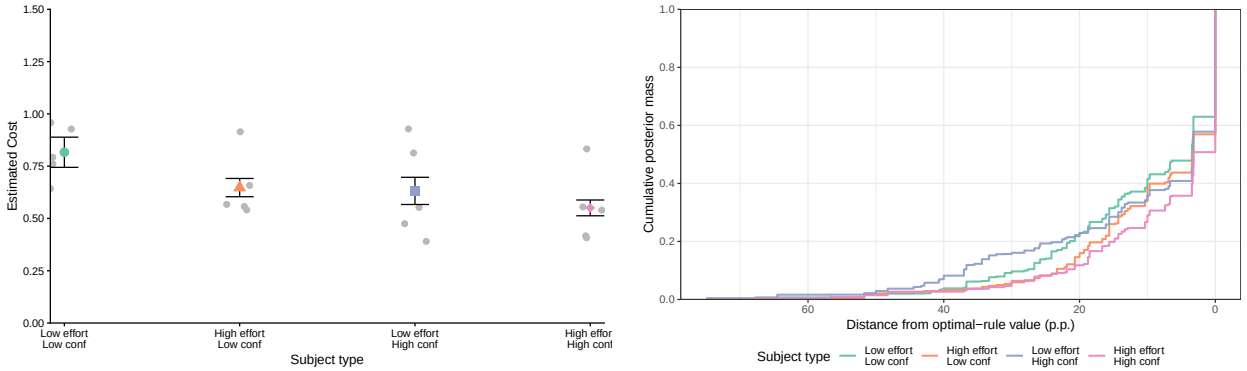
To assess whether the four types indeed capture differences in costs, Figure 4 provides two complementary pieces of evidence.[42] Panel 4a reports the costs implied by subjects' posterior rule assignments, using the identification strategy in Appendix D(iv). Panel 4b plots the cumulative distribution of the accuracy gap between extracted rules and the highest-value rule for each type. At each gap, the curve shows the share of posterior probability assigned to deterministic rules whose value lies at least that distance below the highest-value rule.

Panel 4a supports our interpretation of the four types as proxies for differences in costs. Relative to subjects who find the task *Too Hard*, estimated costs are lower for those who find it *Hard* ($p = 0.045$), *Easy* ($p = 0.058$), or *Doable* ($p = 0.004$). Panel 4b shows how these differences translate into the accuracy of the rules subjects extract. Relative to subjects who find the task *Too Hard*, those who find it *Hard* or *Doable* place more probability on rules closer in accuracy to the highest-value one, in the sense of first-order stochastic dominance: at any given distance from the value-maximizing rule, they place more probability on rules that are at least that close.[43] This result is particularly informative because the effort–confidence classification is only a proxy for unobserved costs, and because first-order

[41] Proposition 5 in Online Appendix OD formalizes these comparative statics, showing that extraction varies continuously with value and that lower-cost agents respond more strongly to differences in rule value, concentrating their choices more heavily on higher-value rules.

[42] The analysis pools the two- and three-light experiments. Results by experiment are reported in Appendix B.

[43] A pooled permutation test rejects equality across the four distributions ($p = 0.008$). Pairwise comparisons show that estimated costs are significantly lower for the *Doable* and *Hard* types than for the *Too Hard* type ($p = 0.004$ and $p = 0.045$, respectively). The *Easy* and *Too Hard* distributions cross: subjects who find the task easy choose higher-value rules on average, but also place more weight on some low-value alternatives, suggesting that perceived ease can reflect either rapid identification or overconfidence when extraction fails.



(a) Estimated costs

(b) CDF of rule extraction by value distance from the highest-value rule

Figure 4: Estimated costs and rule-value distributions across subject types. Panel (a) reports structurally estimated costs recovered from posterior rule assignments; error bars show 95% subject-level bootstrap confidence intervals. Panel (b) reports the posterior-weighted cumulative distribution of extracted rules by value distance from the highest-value rule. Both panels pool observations from the two- and three-light experiments. Stochastic rules are excluded, and the remaining posterior mass is normalized to one within each subject type.

stochastic dominance is stronger than a difference in average rule quality: it orders the entire distribution of extracted rule values.

A complementary prediction is that lower-cost subjects should consider fewer alternative rules: greater sensitivity to value concentrates choice on the more attractive rules, while higher-cost subjects spread probability over a broader set of alternatives. Assigning each subject in each dataset to the rule with the highest posterior probability, the average number of distinct rules used across conditions and experiments falls from 5.65 among subjects who find the task *Too Hard* to 5.05 for *Hard*, 4.90 for *Easy*, and 4.25 for *Doable*. Together, these results provide empirical counterparts to points (iii) and (i) of Proposition 4: lower-cost subjects shift choice toward higher-value rules and concentrate their choices on a smaller set of alternatives.

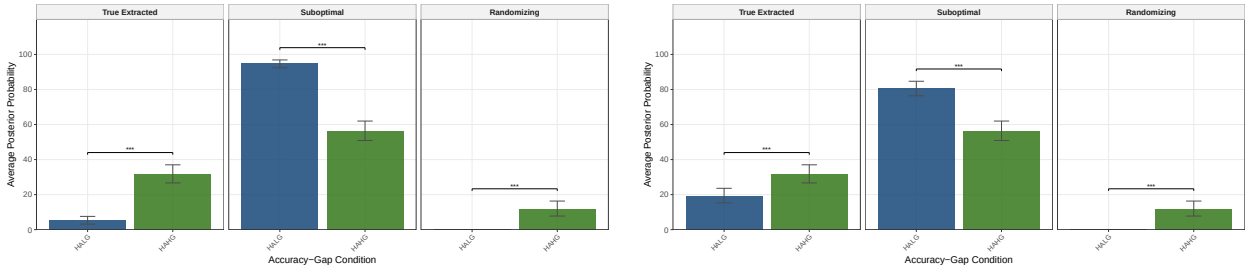
VI(ii) WHEN DO SUBJECTS OVERINFER?

When subjects fail to recover the highest-value rule, do they tend to *overinfer*, using variables that are available but unnecessary for prediction, or instead simplify too much and omit variables that matter? Our data show that both errors occur, and which one emerges depends systematically on the relative value of competing rules.[44] When the highest-value rule uses

[44] Related work documents both tendencies. People infer relationships unsupported by the data and condition on irrelevant variables (Matute et al., 2015; Fréchette et al., 2024), but also favor parsimonious causal explanations involving fewer causes (Lombrozo & Vasilyeva, 2017).

only one variable but two-variable alternatives perform almost as well, subjects often overinfer and condition on an irrelevant variable. Conversely, when the highest-value rule requires both variables but one-variable alternatives perform well, subjects often underinfer and omit a relevant variable. Both errors become much less common when the value-maximizing rule is more clearly separated from its alternatives.[45] Thus, subjects do not display a general preference for richer or simpler rules; they gravitate toward high-performing alternatives, which can lead them either to add irrelevant variables or to omit relevant ones.

A clean test of overinference must compare rules of equal value: otherwise, choosing a rule with more variables may simply reflect its better fit to the observed data. The three-light experiment provides exactly this comparison. When the highest-value rule uses two variables and is weakly separated from its alternatives (HALG), several rules that also condition on the irrelevant third variable achieve exactly the same predictive accuracy. Choosing one of these three-variable rules therefore reflects adding an irrelevant variable without any gain in performance. Figure 5 reports the estimated shares. Panel 5a counts only the two-variable rule as extracted, whereas Panel 5b also counts the equally accurate three-variable rules. The difference between the panels therefore isolates overinference among value-equivalent rules.



(a) Highest-value rule

(b) Value-equivalence class

Figure 5: Posterior rule assignments in the three-light experiment. Panel (a) classifies only the two-light highest-value rule as extracted. Panel (b) also classifies as extracted rules that use the irrelevant third light but attain the same in-sample value as the highest-value rule.

The comparison reveals substantial overinference even among rules that perform equally well. Under the baseline classification, the posterior probability assigned to the highest-value rule is 31.8 percent when it is clearly separated from its alternatives (HAHG), but only 5.2 percent when competing rules perform almost as well (HALG). Once we also count rules that achieve exactly the same in-sample accuracy as the highest-value one, but using 3 variables, the latter rate rises to 19.4 percent, implying that 14.2 percent of posterior mass is assigned

[45] When the highest-value rule uses one variable, widening the gap to the best two-variable alternative from 3.4 to 15.6 percentage points lowers their estimated share from 47.9% to 30.1% ($p = 0.088$); conversely, making one-variable alternatives less competitive lowers their shares from 43.9% to 23.1% ($p = 0.035$) and from 63.7% to 14.0% ($p < 0.001$).

to rules that add variables without improving performance. Taken together, these findings support the stochastic-choice formulation. Even among value-equivalent rules, subjects do not systematically select the one that uses the fewest variables: choices remain dispersed across rules that differ in the number of variables they use. At the same time, overinference is not an unconditional tendency to adopt richer rules. Whether subjects add or omit variables depends on how the resulting rules perform relative to the available alternatives.

VII EMPIRICAL ANALYSIS: KENDALL AND OPREA (2024) DATA

In this section, we reanalyze the data from Kendall and Oprea (2024) with two goals. First, we show that their findings are consistent with the main predictions of our theoretical framework. Second, we use our cost–benefit framework to identify which objective measures of a rule's complexity can be interpreted as arguments of the cost function.

VII(i) THEORETICAL BACKGROUND AND EXPERIMENTAL DESIGN

Kendall and Oprea (2024) study how people learn prediction rules from observations that unfold over time. Each dataset is generated by a finite-state automaton that maps the history of inputs and outputs into the next output. Despite this time dimension, the setting maps naturally into our framework. In our terminology, the automaton generating the output is the (deterministic) DGP, the current input and observed history are the variables, the next output is the outcome to be predicted, a model partitions possible histories into states, and automata are rules that assign a prediction to each state. [46] Because outcomes are generated deterministically, the highest-value automaton coincides with the DGP.

In Part 1 of each task, subjects observe a sequence of twelve inputs, each taking the value a or b and drawn independently with equal probability. After each input is displayed, an output, either x or y, appears, generated by one of eight deterministic data-generating processes reported in Table C1 in Appendix C. Subjects are informed that inputs are exogenous and that outputs are generated by a non-random rule that may depend on the history. In Part 2, subjects observe an additional sequence of twelve inputs and, after each input, are asked to predict the subsequent output by choosing x or y. No feedback is provided until the end of the experiment, when one task is randomly selected for payment, and the subject receives \$0.35 for each correctly predicted output.

Each subject completes eight tasks, each corresponding to a dataset that can be rationalized by a unique DGP.[47] Because each DGP is deterministic, the value-maximizing prediction rule fits every observation in the learning dataset perfectly. For each DGP, however, subjects

[46] Appendix C provides the formal mapping.

[47] We focus on the *Unique* treatment in Kendall and Oprea (2024), which maps naturally into our the-

are randomly assigned one of four datasets, generating exogenous between-subject variation in how well alternative automata fit the data. This design allows us to study how extraction rates vary with the complexity of the DGP and with the particular dataset presented to subjects, affecting the value of alternative lower-value automata.

Main Findings. Kendall and Oprea (2024) show that the probability of extracting the DGP varies systematically with its complexity.[48] Their main result is that extracting the DGP from the data and using it to predict subsequent outcomes is highly nontrivial: extraction rates range from 76 percent for the easiest DGP to 5 percent for the most difficult. Moreover, response times increase sharply with complexity and are negatively associated with extraction success, consistent with complexity capturing automaton-specific costs rather than merely noise or lack of engagement. Several measures of complexity, described in Appendix C, predict this variation, with partition complexity emerging as a particularly strong predictor. Partition complexity captures the number of distinct patterns in the observed history that a decision maker must distinguish to correctly identify and apply an automaton (Lipman, 1995).

VII(ii) OUR ANALYSIS

Based on subjects' choices in Part 2, we extend the analysis of Kendall and Oprea (2024) by inferring which prediction rules (automata) subjects use, rather than simply testing whether they extract the DGP. To parallel our experimental analysis, we encode all 366 automata in the authors' design and estimate posterior probabilities over the full set of candidate automata, providing a richer characterization of rule extraction (Dal Bó & Fréchette, 2011; Dvorak, 2022). Automata that generate identical Part 2 predictions are grouped into observational equivalence classes, since the data cannot distinguish among them.

Table 3 compares the posterior probability assigned to the DGP class with the original extraction measure in Kendall and Oprea (2024), while also showing which alternative automata individuals extract when they fail to recover the DGP. The two approaches yield similar overall estimates of DGP-equivalent extraction: 43.1% under our model and 40.8% under the original classification. The structural estimates reveal what accounts for the remaining behavior, assigning 35.6% of posterior mass to lower-value alternative automata and 21.3% to stochastic behavior that does not condition on the observed history. Moreover,

oretical framework because each dataset is associated with a unique DGP within the candidate class. In the *Multiple* treatment, each dataset is consistent with several two- and three-state DGPs, allowing multiple simple models to rationalize the data.

[48] The DGP is classified as *extracted* if a binomial test rejects, at the 1% level, the null hypothesis that Part 2 guesses are random relative to a data-consistent automaton with partition complexity less than or equal to four. In this design, rejection requires at least 11 correct predictions out of 12.

as the DGP becomes more complex, subjects increasingly substitute toward alternative automata rather than simply randomizing. Interpreting partition complexity as an argument of the cognitive-cost function, this pattern is consistent with our theoretical framework.[49]

DGP Complexity	N	KO(2024)	DGP class	Lower-value Automata	Random
2	297	70.7%	70.9%	3.6%	25.5%
3	198	28.8%	36.4%	42.2%	21.4%
4	99	22.2%	28.0%	43.2%	28.9%
5	99	29.3%	23.0%	57.2%	19.8%
∞	99	5.1%	8.3%	89.6%	2.0%
Overall	792	40.8%	43.1%	35.6%	21.3%

Table 3: Average posterior probabilities by DGP partition complexity. *KO (2024)* reports the original measure of DGP extraction in Kendall and Oprea (2024). The remaining columns report average posterior probabilities from the finite-mixture model. *DGP class* denotes the equivalence group containing the DGP, *Lower-value Automata* denotes all other deterministic automata, and *Random* denotes the stochastic nonconditioning strategy.

Figure 6 reports DGP-extraction rates while holding the partition complexity of the DGP fixed and varying how well the best alternative automaton performs.[50] To test the framework in data not designed around its predictions, we exploit a feature of the experimental design: for each deterministic DGP, subjects are randomly assigned to one of four datasets, generating exogenous variation in how well alternative automata fit the data. Within each DGP-complexity class, we distinguish datasets in which lower-value automata fit the data relatively well from those in which the highest-value automaton is more clearly separated from its alternatives, labeling them *Low Gap* (LG) and *High Gap* (HG), respectively.[51] Subjects extract the DGP more often in *High Gap* datasets, where it stands out more clearly from competing automata, consistent with Proposition 1.

DGP extraction is only part of the picture: when subjects fail to recover the DGP, we also want to know which alternative automata attract them. Table 4 uses posterior probabilities over automata to study this substitution and how it varies with automata values and subjects' costs. Given the available measures, we proxy for these costs using mathematical education.[52]

[49] See Proposition 7 in Online Appendix OD.

[50] As in our experiment, we classify DGP extraction using the exact binomial test at the 1% level.

[51] For each dataset, we compute the accuracy of the best lower-value automaton at each partition-complexity level and average across levels. Within each DGP partition-complexity class, the dataset with the highest average accuracy among these alternatives is classified as *Low Gap*, and the remaining datasets as *High Gap*.

[52] We distinguish subjects with no mathematics coursework, high-school coursework, one university-level course, and two or more university-level courses. Greater mathematical training may lower the cost of identifying and applying an automaton (Fong et al., 1986).

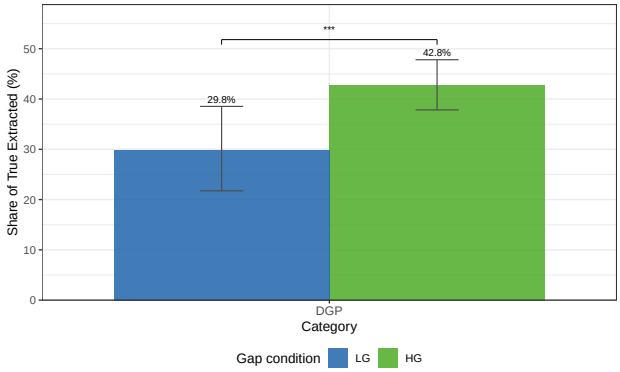


Figure 6: DGP extraction by value gap. Bars report the shares of subject–dataset observations classified as extracting the DGP in the *High Gap* and *Low Gap* conditions. Error bars show 95% subject-level bootstrap confidence intervals.

Outcome	All-automata posterior		Statistical Test Assignment	
Specification	Pooled	Cost Types	True Extracted	Randomizing
	(1)	(2)	(3)	(4)
Constant	24.3***	11.5**	14.9***	70.0***
	(1.17)	(4.65)	(5.49)	(8.46)
Gap condition: LG	-1.87***	-1.75***	-12.7***	10.1**
	(0.405)	(0.426)	(3.92)	(4.87)
Distance (p.p.)	-0.346***	0.027		
	(0.036)	(0.126)		
Math type: High school		9.05*	20.0**	-13.8
		(5.32)	(7.95)	(10.6)
Math type: One university course		10.6**	23.8***	-17.7*
		(4.90)	(6.23)	(9.10)
Math type: Two+ university courses		16.4***	36.0***	-30.2***
		(4.92)	(6.71)	(9.22)
Distance × High school		-0.257*		
		(0.148)		
Distance × One university course		-0.307**		
		(0.136)		
Distance × Two+ university courses		-0.487***		
		(0.136)		
Observations	6,398	6,398	784	784
R^2	0.048	0.054	0.042	0.034

Notes: Standard errors in parentheses are heteroskedasticity-robust and clustered by subject. The *All-strategy posterior* columns use posterior assignment probabilities, so each subject enters once for every retained candidate equivalence class in each dataset; these probabilities sum to one within each subject–dataset pair. The other columns use binary assignments from the exact-test classification. The omitted subject type is No courses, and HG is the omitted condition.
$^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table 4: Extraction rates by automata accuracy and mathematical education: Evidence from Kendall and Oprea (2024)

Column (1) first shows that an automaton's value strongly predicts extraction: each additional prediction it misses relative to the DGP's 12 correct predictions lowers its posterior probability by 2.88 percentage points. Thus, when subjects fail to recover the DGP, they substitute toward the alternative automata that best fit the learning data. Column (2)

shows that this response to value becomes stronger with mathematical education. Relative to subjects with no mathematics coursework, those with greater training place more weight on high-performing automata and reduce that weight more sharply as accuracy deteriorates. This pattern is consistent with Proposition 4: when costs are lower, choices respond more strongly to differences in value and concentrate on better-performing alternatives.

Columns (3) and (4) use the binary classifications from the exact statistical tests underlying the aggregate extraction rates in Figure 6. Controlling for mathematical education leaves the value-gap pattern essentially unchanged: relative to HG, LG lowers DGP extraction by 12.7 percentage points. This reinforces the value-based substitution predicted by Proposition 1, showing that the effect is present across subjects and is not driven by specific types. Mathematical education nevertheless matters for how subjects respond to the task. Relative to subjects with no mathematics coursework, DGP extraction rises systematically with training and is 36 percentage points higher among those with two or more university-level courses, a large and statistically significant difference. As shown in Column (4), the most mathematically trained subjects are also less likely to be classified as randomizing. Together, these patterns are consistent with Proposition 4: greater mathematical training is associated with more frequent DGP extraction and stronger responses to differences in automata performance.

VII(ii).a The argument of the cost function

Kendall and Oprea (2024) compare how strongly several objective complexity measures are associated with DGP extraction and use these relationships to identify which dimensions of complexity best predict learning. In our framework, however, subjects substitute across competing automata, so DGP extraction depends not only on complexity costs but also on how attractive lower-value alternatives are. Raw associations between complexity and extraction may therefore partly reflect value-driven substitution rather than complexity itself. We therefore extend their analysis to the full set of automata and control explicitly for their values. This allows us to separate costs from benefits and assess which objective measures of a rule's complexity can be interpreted as arguments of the cost function after accounting for value-driven substitution.

Figure 7 summarizes the ordinal associations (Goodman–Kruskal γ statistic) between complexity and extraction under three specifications. First, the *Kendall and Oprea, 2024* specification replicates the original benchmark at the subject–dataset level, relating the true DGP's complexity to the binary indicator of DGP extraction obtained from the exact binomial test. The *All Automata* specification extends the analysis to the automata assigned to subjects by the mixture model, still without controlling for value.[53] Finally, *All Automata*

[53] For each subject–dataset pair, we code as extracted the equivalence class with the highest posterior

(conditional) restricts attention to automata that correctly predict at least 8 of the 12 learning observations and estimates the association separately within each exact value level before pooling across levels. This controls for value while avoiding results driven by very low-value automata, which are naturally less likely to be extracted.

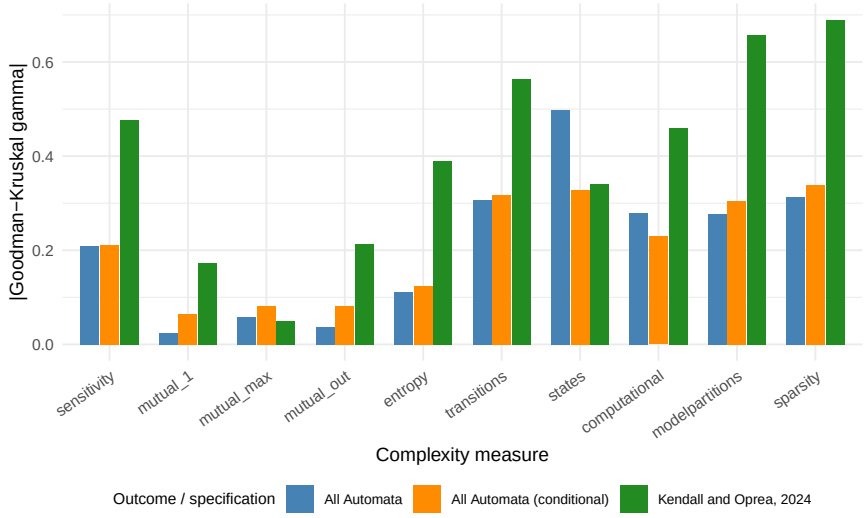


Figure 7: Goodman–Kruskal γ statistic, which ranges from -1 for a perfect negative association to $+1$ for a perfect positive association. The figure reports $|\gamma|$, so taller bars indicate stronger associations regardless of direction. *Kendall and Oprea, 2024* relates true-DGP complexity to DGP extraction. *All Automata* extends the comparison to all extracted automata. *All Automata (conditional)* compares automata holding value fixed, thereby isolating the association between complexity and extraction ceteris paribus with respect to performance.

Two main results emerge. First, the relationship between complexity and extraction documented by Kendall and Oprea (2024) for the DGP extends to the full set of automata: partition and sparsity complexity remain among the strongest predictors of which automata subjects extract. Second, these associations generally weaken once all automata are considered and comparisons are made holding value fixed. The decline is especially pronounced for measures that primarily capture properties of the observed data and can therefore affect the value of alternative automata. Entropy, sensitivity, and current-input mutual information provide clear examples.[54] Relative to the original Kendall–Oprea estimates, conditioning on learning-sample performance substantially reduces their association with extraction,[55] sug-

probability from the structural estimation. Because a class may contain multiple automata with identical test-sample predictions, we represent it by the automaton(s) with the highest learning-sample performance, splitting weight equally across ties so the class receives total weight one. If the automaton prescribing unconditional randomization has the highest posterior, no deterministic automaton is coded as extracted.

[54] Entropy captures output variability, sensitivity measures how strongly the output depends on the history of inputs, and *mutual_1* measures how informative the current input is about the output.

[55] For entropy, $|\gamma|$ falls from 0.390 to 0.124; for sensitivity, from 0.477 to 0.210; and for current-input

gesting that part of their raw predictive power reflects value-driven substitution between the DGP and lower-value automata rather than complexity costs.

By contrast, measures tied more directly to the structural properties of the automata, such as sparsity and partition complexity, remain comparatively strong predictors of extraction. The number of states, transitions, and computational complexity, which were less prominent in the raw associations, also become relatively more important once all automata are considered and differences in value are held fixed. This pattern is consistent with costs arising from the representation, memory, and implementation demands of the automata (Oprea, 2026). Overall, the figure draws a clear distinction: measures describing statistical properties of the data weaken substantially once value is held fixed, whereas measures describing the structure of the automata retain strong associations with extraction.

VIII CONCLUSION

Learning often involves more than updating beliefs as information accumulates. In many settings, people must infer relationships directly from data, without prior information. A central lesson is that relationships learned from one environment may fit the observed data well yet fail to capture the stable mechanism that governs outcomes across environments.

In this paper, we propose the first theoretical framework to formalize how agents extract behavioral rules directly from data in the absence of prior information. We characterize how the values and complexity costs of competing rules jointly determine whether agents recover the highest-value rule that generalizes across environments or instead substitute toward alternatives that perform well only in particular datasets. We then show how these extraction patterns vary with the statistical structure of the data, incentives, the number of variables, and heterogeneity in cognitive costs. A central implication is that agents do not extract the generalizable rule simply because it attains the highest value; extraction also depends on how clearly it outperforms alternative rules. Costs, in turn, determine how strongly agents respond to these differences in value: lower-cost agents extract the highest-value rule more often, randomize less, and, when they make mistakes, select rules closer to the value-maximizing one.

Separating rule value from cognitive cost also changes how complexity should be interpreted empirically. Reanalyzing the data from Kendall and Oprea (2024) through our framework allows us to separate complexity-driven effects from value-driven substitution across competing rules. Once value is held fixed, structural features of automata—including states, transitions, sparsity, partitions, and computational requirements—remain comparatively strong predictors of rule extraction, whereas associations with statistical properties

mutual information, from 0.174 to 0.064.

of the dataset, such as entropy and sensitivity to past histories, weaken substantially. This pattern is consistent with structural features of rules, rather than statistical properties of datasets, being the relevant sources of complexity costs.

More broadly, the framework opens a research agenda on how observable features of the environment, such as the statistical properties of the dataset, shape the behavioral rules people extract. While the datasets are exogenous in this paper, a natural next step is to endogenize them, allowing agents' actions to shape the observations from which they subsequently learn and creating an equilibrium feedback between behavior and data. Moreover, the framework can organize familiar behavioral biases as endogenous responses to features of the environment rather than as primitive psychological dispositions. For example, confounder neglect (Spiegler, 2016, 2020), conditioning on irrelevant variables (Fréchette et al., 2024), failure to use relevant predictors (Hanna et al., 2014), coarse thinking (Jehiel, 2005; Mullainathan et al., 2008), and neglect of indirect effects (Spiegler, 2020) should each become more prevalent when the corresponding misspecified rule performs well relative to competing alternatives. The framework thus moves from documenting behavioral biases to predicting which biases should emerge in which environments.

Appendix

A PROOFS

In what follows we adopt the following notation. First, we define the "total salience" for a set $\hat{\Pi} \subseteq \Pi$ as $\kappa(\hat{\Pi}) = \sum_{\pi_M \in \hat{\Pi}} \frac{1}{\kappa(\pi_M)}$. Second, we define the average weighted value as $v^{\mathcal{D}}(\hat{\Pi}) = \frac{\sum_{\pi_M \in \hat{\Pi}} \frac{1}{\kappa(\pi_M)} v^{\mathcal{D}}(\pi_M)}{\sum_{\pi_M \in \hat{\Pi}} \frac{1}{\kappa(\pi_M)}}$.

Without loss of generality, we also assume that all rules attain strictly positive values. This assumption is without loss of generality because adding a common constant $b \in \mathbb{R}$ to all rule values leaves choice probabilities unchanged: the numerator of the comparative value term, $v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\hat{\Pi})$, is invariant under such a shift, as is the difference $v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})$ that determines choice probabilities, since both $v^{\mathcal{D}}(\pi_M)$ and $\Lambda(\Pi^{\mathcal{D}})$ shift by the same constant b.

Finally, denote by $\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') := \frac{v^{\mathcal{D}'}(\pi_M)}{v^{\mathcal{D}}(\pi_M)}$ the ratio of the value of rule π_M across decision problems $\mathcal{D}$ and $\mathcal{D}'$.

A(i) ALTERNATIVE INTERPRETATION OF THE MAXIMIZATION PROBLEM

Because agents have no prior information and must extract all payoff-relevant information from the data, we take random choice as the natural benchmark. Let $\pi^R \in \Pi$ denote the rule that ignores all predictors and randomizes over actions, and assume that it has the lowest cost: $\kappa^R \equiv \kappa(\pi^R) = \min_{\pi_M \in \Pi} \kappa(\pi_M)$. For each decision problem $\mathcal{D}$, define the value of a rule relative to random choice as $\widetilde{v}^{\mathcal{D}}(\pi_M) = v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi^R)$, so that $\widetilde{v}^{\mathcal{D}}(\pi^R) = 0$. Because extraction probabilities sum to one, replacing $v^{\mathcal{D}}$ with $\widetilde{v}^{\mathcal{D}}$ changes the objective in equation (1) only by the constant $v^{\mathcal{D}}(\pi^R)$. The cost can likewise be normalized relative to assigning probability one to the random rule and equation (1) can therefore be written equivalently as

$$\rho^{\mathcal{D}}(\cdot) = \arg\max_{\rho \in \Delta(\Pi)} \left\{ \sum_{\pi_M \in \Pi} \widetilde{v}^{\mathcal{D}}(\pi_M) \rho(\pi_M) - \frac{1}{2} \sum_{\pi_M \in \Pi} \kappa(\pi_M) \rho(\pi_M)^2 + \frac{\kappa^R}{2} \right\}.$$

Because the normalization changes the objective only by constants, it leaves the solution unchanged. It makes explicit that the agent trades off improvements in performance relative to random choice against the costs of extracting alternative rules.

A(ii) SOLUTION TO THE MAXIMIZATION PROBLEM

A necessary and sufficient condition for $\rho^{\mathcal{D}}(\pi_M) > 0$ for all $\pi_M \in \Pi$ is $v^{\mathcal{D}}(\pi_M) > \Lambda(\Pi) := v^{\mathcal{D}}(\Pi) - \frac{1}{\sum_{\pi_M \in \Pi} \frac{1}{\kappa(\pi_M)}}$.[56] This condition requires the value of all rules to be larger than the shadow price of relaxing the constraint that probabilities sum to 1 in the choice set Π. However, it could be the case that some rules are chosen with zero probability. To account for that, we define $\Pi^{\mathcal{D}}$ to be the unique subset of Π satisfying $\Pi^{\mathcal{D}} = \{\pi_M \in \Pi : v^{\mathcal{D}}(\pi_M) > \Lambda(\Pi^{\mathcal{D}})\}$.[57] Note that $\pi^{\max}_{M^*} \in \Pi^{\mathcal{D}}$. The probability of choosing rule π_M can then be written as:

$$\rho^{\mathcal{D}}(\pi_M) = \begin{cases} \frac{\frac{1}{\kappa(\pi_M)}}{\sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{1}{\kappa(\pi_M)}} + \frac{1}{\kappa(\pi_M)} \left[v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\Pi^{\mathcal{D}}) \right] & \pi_M \in \Pi^{\mathcal{D}} \\ 0 & \pi_M \notin \Pi^{\mathcal{D}} \end{cases}$$

A(iii) PROOFS OF THE MAIN PROPOSITIONS

A(iii).a Proof of Proposition 1

For every $\pi_M \in \Pi^{\mathcal{D}}$, we can write

$$\rho^{\mathcal{D}}(\pi_M) = \left(v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}}) \right) \frac{1}{\kappa(\pi_M)}. \tag{4}$$

First, we prove that the highest-value rule is extracted with positive probability. Since $\Pi^{\mathcal{D}}$ is nonempty,

$$\Lambda(\Pi^{\mathcal{D}}) = v^{\mathcal{D}}(\Pi^{\mathcal{D}}) - \frac{1}{\kappa(\Pi^{\mathcal{D}})} < \max_{\pi_M \in \Pi^{\mathcal{D}}} v^{\mathcal{D}}(\pi_M) \leq v^{\mathcal{D}}(\pi^{\max}_{M^*}).$$

It follows from the definition of the consideration set that $\pi^{\max}_{M^*} \in \Pi^{\mathcal{D}}$. Equation (4) therefore gives

$$\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) = \left(v^{\mathcal{D}}(\pi^{\max}_{M^*}) - \Lambda(\Pi^{\mathcal{D}}) \right) \frac{1}{\kappa(\pi^{\max}_{M^*})} > 0.$$

If $\Pi \setminus \{\pi^{\max}_{M^*}\} = \varnothing$, the remaining claims are immediate. Hence, suppose that $\Pi \setminus \{\pi^{\max}_{M^*}\} \neq \varnothing$ and denote by

$$\alpha^{\max} := \max_{\pi_M \in \Pi \setminus \{\pi^{\max}_{M^*}\}} \left(\Delta^{\pi_M}(\mathcal{D}, \mathcal{D}') - 1 \right) v^{\mathcal{D}}(\pi_M).$$

Applying the preceding argument to both decision problems shows that the highest-value rule belongs to both consideration sets. We can therefore write the difference in its extraction

[56] The proof follows the same steps as the proof of Proposition 1 in Online Appendix A1 of Chambers et al. (2025), with the appropriate notational changes.

[57] Proposition 5 in Online Appendix B1 of Chambers et al., 2025 provides an algorithm showing how to construct the set $\Pi^{\mathcal{D}}$ starting from the primitives u, κ and Π. The proposition also shows that the set is unique and never empty.

probabilities as

$$\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) - \rho^{\mathcal{D}}(\pi^{\max}_{M^*}) = \frac{1}{\kappa(\pi^{\max}_{M^*})}\Big[\Big(\Delta^{\pi^{\max}_{M^*}}(\mathcal{D},\mathcal{D}') - 1\Big)\, v^{\mathcal{D}}(\pi^{\max}_{M^*}) - \Big(\Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}})\Big)\Big]. \tag{5}$$

Since $\kappa(\pi^{\max}_{M^*}) > 0$, this difference is positive if and only if

$$\Big(\Delta^{\pi^{\max}_{M^*}}(\mathcal{D},\mathcal{D}') - 1\Big)\, v^{\mathcal{D}}(\pi^{\max}_{M^*}) > \Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}}). \tag{6}$$

The value-gap condition implies

$$\Big(\Delta^{\pi^{\max}_{M^*}}(\mathcal{D},\mathcal{D}') - 1\Big)\, v^{\mathcal{D}}(\pi^{\max}_{M^*}) > \alpha^{\max}.$$

We first establish the weak inequality. Suppose, toward a contradiction, that $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) < \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$. Equation (5) then implies

$$\Big(\Delta^{\pi^{\max}_{M^*}}(\mathcal{D},\mathcal{D}') - 1\Big)\, v^{\mathcal{D}}(\pi^{\max}_{M^*}) < \Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}}),$$

and hence $\alpha^{\max} < \Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}})$. Moreover, it must be that $\Pi^{\mathcal{D}'} \subseteq \Pi^{\mathcal{D}}$. Indeed, if there existed some $\pi_M \in \Pi^{\mathcal{D}'} \setminus \Pi^{\mathcal{D}}$, then

$$v^{\mathcal{D}'}(\pi_M) > \Lambda(\Pi^{\mathcal{D}'}) \qquad \text{and} \qquad v^{\mathcal{D}}(\pi_M) \leq \Lambda(\Pi^{\mathcal{D}}),$$

so that

$$\left(\Delta^{\pi_M}(\mathcal{D},\mathcal{D}') - 1\right) v^{\mathcal{D}}(\pi_M) > \Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}}).$$

Since such a rule is different from $\pi^{\max}_{M^*}$, the left-hand side is at most $\alpha^{\max}$, which is a contradiction.

For every $\pi_M \in \Pi^{\mathcal{D}} \cap \Pi^{\mathcal{D}'}$, equation (4) gives

$$\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M) = \frac{1}{\kappa(\pi_M)}\Big[\left(\Delta^{\pi_M}(\mathcal{D},\mathcal{D}') - 1\right) v^{\mathcal{D}}(\pi_M) - \Big(\Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}})\Big)\Big] < 0.$$

For the highest-value rule, the strict inequality follows from the contradiction hypothesis; for every other rule, it follows from the definition of $\alpha^{\max}$. If instead $\pi_M \in \Pi^{\mathcal{D}} \setminus \Pi^{\mathcal{D}'}$, then $\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M) = -\rho^{\mathcal{D}}(\pi_M) < 0$. Thus no new rule enters and every rule that initially receives positive probability loses probability, contradicting $\sum_{\pi_M \in \Pi}\Big[\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M)\Big] = 0$. Therefore, $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \geq \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$.

Finally, suppose that $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$ and, toward a contradiction, that $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) = \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$. Equation (5) then implies

$$\Big(\Delta^{\pi^{\max}_{M^*}}(\mathcal{D},\mathcal{D}') - 1\Big)\, v^{\mathcal{D}}(\pi^{\max}_{M^*}) = \Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}}),$$

and therefore $\alpha^{\max} < \Lambda(\Pi^{\mathcal{D}'}) - \Lambda(\Pi^{\mathcal{D}})$. The same argument as above implies that $\Pi^{\mathcal{D}'} \subseteq \Pi^{\mathcal{D}}$ and that every $\pi_M \in \Pi^{\mathcal{D}} \setminus \{\pi^{\max}_{M^*}\}$ loses probability strictly. Since $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$, at least one such rule receives positive probability under $\mathcal{D}$. The probability of the highest-value rule is unchanged, no new rule enters, and at least one alternative loses probability strictly.

This again contradicts the requirement that probabilities sum to one. Hence, $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) > \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$ whenever $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$.

A(iv) Proof of Corollary 1.

By linearity of rule values in utility,

$$v^{\mathcal{D}'}(\pi_M) = \alpha v^{\mathcal{D}}(\pi_M) \qquad \text{for every } \pi_M \in \Pi. \tag{7}$$

Since $\pi^{\max}_{M^*}$ is the unique value-maximizing rule in $\mathcal{D}$, it is also the unique value-maximizing rule in $\mathcal{D}'$. Moreover, for every $\pi_M \in \Pi \setminus \{\pi^{\max}_{M^*}\}$,

$$v^{\mathcal{D}'}(\pi^{\max}_{M^*}) - v^{\mathcal{D}'}(\pi_M) = \alpha \left[v^{\mathcal{D}}(\pi^{\max}_{M^*}) - v^{\mathcal{D}}(\pi_M) \right].$$

If $\alpha > 1$, every value gap is therefore strictly larger under $\mathcal{D}'$ than under $\mathcal{D}$. Proposition 1 then implies $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \geq \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$, with strict inequality if $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$.

If $\alpha \in (0, 1)$, every value gap is strictly larger under $\mathcal{D}$ than under $\mathcal{D}'$. Applying Proposition 1 with $\mathcal{D}$ and $\mathcal{D}'$ interchanged gives $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \leq \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$. If $\rho^{\mathcal{D}}(\pi^{\max}_{M^*}) < 1$, this weak inequality implies $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) < 1$. The strict part of Proposition 1, again applied with the two decision problems interchanged, therefore implies strict inequality.

We next prove the claims about the consideration sets. Recall that a rule π_M belongs to $\Pi^{\mathcal{D}}$ if and only if $v^{\mathcal{D}}(\pi_M) > \Lambda(\Pi^{\mathcal{D}})$. By equation (7), a rule π_M belongs to $\Pi^{\mathcal{D}'}$ if and only if

$$v^{\mathcal{D}}(\pi_M) > v^{\mathcal{D}}(\Pi^{\mathcal{D}'}) - \frac{1}{\kappa(\Pi^{\mathcal{D}'})\alpha} = \frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha}. \tag{8}$$

Using equation (4) and the fact that extraction probabilities sum to one, we have

$$1 = \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)},$$

and, after dividing the corresponding equality for $\mathcal{D}'$ by α,

$$\frac{1}{\alpha} = \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}}(\pi_M) - \frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha}}{\kappa(\pi_M)}.$$

Suppose first that $\alpha > 1$ and, toward a contradiction, that

$$\frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha} \leq \Lambda(\Pi^{\mathcal{D}}).$$

Equation (8) then implies $\Pi^{\mathcal{D}} \subseteq \Pi^{\mathcal{D}'}$, and hence

$$\begin{aligned} \frac{1}{\alpha} &= \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}}(\pi_M) - \frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha}}{\kappa(\pi_M)} \geq \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha}}{\kappa(\pi_M)} \\ &\geq \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} = 1, \end{aligned}$$

which contradicts $\alpha > 1$. Therefore,

$$\frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha} > \Lambda(\Pi^{\mathcal{D}}),$$

and equation (8) implies $\Pi^{\mathcal{D}'} \subseteq \Pi^{\mathcal{D}}$.

Suppose instead that $\alpha \in (0,1)$ and, toward a contradiction, that

$$\frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha} \geq \Lambda(\Pi^{\mathcal{D}}).$$

Equation (8) then implies $\Pi^{\mathcal{D}'} \subseteq \Pi^{\mathcal{D}}$, and hence

$$\begin{aligned}\frac{1}{\alpha} &= \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}}(\pi_M) - \frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha}}{\kappa(\pi_M)} \leq \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} \\ &\leq \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} = 1,\end{aligned}$$

which contradicts $\alpha \in (0,1)$. Therefore,

$$\frac{\Lambda(\Pi^{\mathcal{D}'})}{\alpha} < \Lambda(\Pi^{\mathcal{D}}),$$

and equation (8) implies $\Pi^{\mathcal{D}'} \supseteq \Pi^{\mathcal{D}}$.

Finally, consider the limits. Since Π is finite and $\pi^{\max}_{M^*}$ is the unique value-maximizing rule, for all sufficiently large α,

$$\alpha \left[v^{\mathcal{D}}(\pi^{\max}_{M^*}) - v^{\mathcal{D}}(\pi_M) \right] > \kappa(\pi^{\max}_{M^*}) \qquad \text{for every } \pi_M \in \Pi \setminus \{\pi^{\max}_{M^*}\}.$$

For such α, the cutoff associated with the singleton $\{\pi^{\max}_{M^*}\}$ is $\alpha v^{\mathcal{D}}(\pi^{\max}_{M^*}) - \kappa(\pi^{\max}_{M^*})$. The highest-value rule lies strictly above this cutoff, whereas every other rule lies strictly below it. Hence, by uniqueness of the consideration set, $\Pi^{\mathcal{D}'} = \{\pi^{\max}_{M^*}\}$ for all sufficiently large α. Consequently, $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) = 1$ for all sufficiently large α, and therefore $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \to 1$ as $\alpha \to \infty$.

As $\alpha \downarrow 0$, finiteness of Π implies that, for all sufficiently small α,

$$\alpha \left[v^{\mathcal{D}}(\Pi) - v^{\mathcal{D}}(\pi_M) \right] < \frac{1}{\kappa(\Pi)} \qquad \text{for every } \pi_M \in \Pi.$$

Equivalently,

$$v^{\mathcal{D}'}(\pi_M) > v^{\mathcal{D}'}(\Pi) - \frac{1}{\kappa(\Pi)} \qquad \text{for every } \pi_M \in \Pi.$$

Thus, by uniqueness of the consideration set, $\Pi^{\mathcal{D}'} = \Pi$ for all sufficiently small α. Equation (2) then gives

$$\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) = \frac{\frac{1}{\kappa(\pi^{\max}_{M^*})}}{\sum_{\pi_M \in \Pi} \frac{1}{\kappa(\pi_M)}} + \frac{\alpha}{\kappa(\pi^{\max}_{M^*})} \left[v^{\mathcal{D}}(\pi^{\max}_{M^*}) - v^{\mathcal{D}}(\Pi) \right].$$

Taking $\alpha \downarrow 0$ yields

$$\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) \to \frac{\frac{1}{\kappa(\pi^{\max}_{M^*})}}{\sum_{\pi_M \in \Pi} \frac{1}{\kappa(\pi_M)}}.$$

A(v) Proof of Proposition 2.

Since $\mathcal{D}$ and $\mathcal{D}'$ are equivalent except for rule values, they have the same candidate rule set Π and the same cost function $\kappa(\cdot)$. Recall that

$$\rho^{\mathcal{D}}(\pi_M) = \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} \qquad \text{for every } \pi_M \in \Pi^{\mathcal{D}},$$

and similarly for $\mathcal{D}'$. Since $\rho^{\mathcal{D}}(\pi^R) > 0$, we have

$$\pi^R \in \Pi^{\mathcal{D}} \qquad \text{and} \qquad v^{\mathcal{D}}(\pi^R) > \Lambda(\Pi^{\mathcal{D}}).$$

Moreover, since $\rho^{\mathcal{D}}(\pi^R) < 1$, there exists some $\widehat{\pi}_{\widehat{M}} \in \Pi^{\mathcal{D}} \setminus \{\pi^R\}$ such that $\rho^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}}) > 0$.
We first show that $\Lambda(\Pi^{\mathcal{D}'}) < \Lambda(\Pi^{\mathcal{D}})$. Suppose, toward a contradiction, that $\Lambda(\Pi^{\mathcal{D}'}) \geq \Lambda(\Pi^{\mathcal{D}})$. This implies that $\Pi^{\mathcal{D}'} \subseteq \Pi^{\mathcal{D}}$. Indeed, if $\pi_M \in \Pi^{\mathcal{D}'}$, then $v^{\mathcal{D}'}(\pi_M) > \Lambda(\Pi^{\mathcal{D}'}) \geq \Lambda(\Pi^{\mathcal{D}})$. If $\pi_M = \pi^R$, then $v^{\mathcal{D}}(\pi^R) = v^{\mathcal{D}'}(\pi^R)$, whereas if $\pi_M \neq \pi^R$, then $v^{\mathcal{D}}(\pi_M) > v^{\mathcal{D}'}(\pi_M)$. In either case, $v^{\mathcal{D}}(\pi_M) > \Lambda(\Pi^{\mathcal{D}})$, and hence $\pi_M \in \Pi^{\mathcal{D}}$.

Using the fact that extraction probabilities sum to one, we therefore obtain

$$\begin{aligned} 1 &= \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}'}(\pi_M) - \Lambda(\Pi^{\mathcal{D}'})}{\kappa(\pi_M)} \leq \sum_{\pi_M \in \Pi^{\mathcal{D}'}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} \\ &\leq \sum_{\pi_M \in \Pi^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi_M)} = 1. \end{aligned}$$

At least one of the two inequalities is strict. If $\widehat{\pi}_{\widehat{M}} \in \Pi^{\mathcal{D}'}$, the first inequality is strict because $v^{\mathcal{D}'}(\widehat{\pi}_{\widehat{M}}) < v^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}})$. If instead $\widehat{\pi}_{\widehat{M}} \notin \Pi^{\mathcal{D}'}$, the second inequality is strict because $v^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}}) - \Lambda(\Pi^{\mathcal{D}}) > 0$. This yields a contradiction. Therefore, $\Lambda(\Pi^{\mathcal{D}'}) < \Lambda(\Pi^{\mathcal{D}})$.

Finally,

$$v^{\mathcal{D}'}(\pi^R) = v^{\mathcal{D}}(\pi^R) > \Lambda(\Pi^{\mathcal{D}}) > \Lambda(\Pi^{\mathcal{D}'}),$$

so that $\pi^R \in \Pi^{\mathcal{D}'}$. It follows that

$$\begin{aligned} \rho^{\mathcal{D}'}(\pi^R) - \rho^{\mathcal{D}}(\pi^R) &= \frac{v^{\mathcal{D}'}(\pi^R) - \Lambda(\Pi^{\mathcal{D}'})}{\kappa(\pi^R)} - \frac{v^{\mathcal{D}}(\pi^R) - \Lambda(\Pi^{\mathcal{D}})}{\kappa(\pi^R)} \\ &= \frac{\Lambda(\Pi^{\mathcal{D}}) - \Lambda(\Pi^{\mathcal{D}'})}{\kappa(\pi^R)} > 0. \end{aligned}$$

Hence, $\rho^{\mathcal{D}'}(\pi^R) > \rho^{\mathcal{D}}(\pi^R)$.

A(v).a Proof of Proposition 3

Let $\Pi(\mathcal{D})$ denote the full set of admissible rules in problem $\mathcal{D}$. Since $N_{\mathcal{D}} \subset N_{\mathcal{D}'}$ and $\text{proj}_{N_{\mathcal{D}}} X_{N_{\mathcal{D}'}} = X_{N_{\mathcal{D}}}$, we have $\Pi(\mathcal{D}) \subset \Pi(\mathcal{D}')$. We consider two cases.

Case 1: No new rule enters the consideration set. If $v^{\mathcal{D}'}(\pi'_{M'}) \leq \Lambda(\Pi^{\mathcal{D}})$ for every $\pi'_{M'} \in \Pi(\mathcal{D}') \setminus \Pi(\mathcal{D})$, then $\Pi^{\mathcal{D}'} = \Pi^{\mathcal{D}}$ and, since rule values are unchanged, $\rho^{\mathcal{D}'}(\pi_M) = \rho^{\mathcal{D}}(\pi_M)$ for all $\pi_M \in \Pi^{\mathcal{D}}$.

Case 2: At least one new rule enters the consideration set. Suppose there exists $\pi'_{M'} \in \Pi(\mathcal{D}') \setminus \Pi(\mathcal{D})$ such that $v^{\mathcal{D}'}(\pi'_{M'}) > \Lambda(\Pi^{\mathcal{D}})$. Then, by Lemma 1 in the Online Appendix of Chambers et al. (2025), it follows that $\Lambda(\Pi^{\mathcal{D}'}) > \Lambda(\Pi^{\mathcal{D}})$. For any $\pi_M \in \Pi^{\mathcal{D}}$, two sub-cases apply. If $\pi_M \in \Pi^{\mathcal{D}'}$, then since $v^{\mathcal{D}'}(\pi_M) = v^{\mathcal{D}}(\pi_M)$, equation (4) gives $\rho^{\mathcal{D}'}(\pi_M) = \frac{1}{\kappa(\pi_M)}(v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}'})) < \frac{1}{\kappa(\pi_M)}(v^{\mathcal{D}}(\pi_M) - \Lambda(\Pi^{\mathcal{D}})) = \rho^{\mathcal{D}}(\pi_M)$. If $\pi_M \notin \Pi^{\mathcal{D}'}$, then $\rho^{\mathcal{D}'}(\pi_M) = 0 < \rho^{\mathcal{D}}(\pi_M)$. In both sub-cases $\rho^{\mathcal{D}'}(\pi_M) < \rho^{\mathcal{D}}(\pi_M)$ for all $\pi_M \in \Pi^{\mathcal{D}}$, and in particular $\rho^{\mathcal{D}'}(\pi^{\max}_{M^*}) < \rho^{\mathcal{D}}(\pi^{\max}_{M^*})$.

A(v).b Proof of Proposition 4

Both types face the same decision problem $\mathcal{D}$ and hence the same rule values $v^{\mathcal{D}}(\cdot)$. For ease of notation, write $\Lambda_i^{\mathcal{D}} := \Lambda(\Pi_i^{\mathcal{D}}), \qquad i \in \{1, 2\}$. Equation (4) and the definition of the consideration set give

$$\rho_i^{\mathcal{D}}(\pi_M) = \begin{cases} \dfrac{v^{\mathcal{D}}(\pi_M) - \Lambda_i^{\mathcal{D}}}{\kappa_i(\pi_M)}, & \pi_M \in \Pi_i^{\mathcal{D}}, \\ 0, & \pi_M \notin \Pi_i^{\mathcal{D}}, \end{cases}$$

where

$$\Pi_i^{\mathcal{D}} = \left\{ \pi_M \in \Pi : v^{\mathcal{D}}(\pi_M) > \Lambda_i^{\mathcal{D}} \right\}.$$

Since $\rho_i^{\mathcal{D}}$ is a probability distribution,

$$1 = \sum_{\pi_M \in \Pi_i^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_i^{\mathcal{D}}}{\kappa_i(\pi_M)}.$$

Step 1: Ordering of the cutoffs and consideration sets.

Suppose, toward a contradiction, that $\Lambda_2^{\mathcal{D}} \leq \Lambda_1^{\mathcal{D}}$. The definition of the consideration sets then implies $\Pi_1^{\mathcal{D}} \subseteq \Pi_2^{\mathcal{D}}$. Using $\kappa_1(\cdot) = \gamma\kappa_2(\cdot)$ and the normalization conditions,

$$\begin{aligned} 1 &= \sum_{\pi_M \in \Pi_2^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_2^{\mathcal{D}}}{\kappa_2(\pi_M)} \geq \sum_{\pi_M \in \Pi_1^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_2^{\mathcal{D}}}{\kappa_2(\pi_M)} \\ &\geq \sum_{\pi_M \in \Pi_1^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_1^{\mathcal{D}}}{\kappa_2(\pi_M)} = \gamma \sum_{\pi_M \in \Pi_1^{\mathcal{D}}} \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_1^{\mathcal{D}}}{\kappa_1(\pi_M)} = \gamma, \end{aligned}$$

which contradicts $\gamma > 1$. Therefore, $\Lambda_2^{\mathcal{D}} > \Lambda_1^{\mathcal{D}}$. It follows immediately that

$$\Pi_2^{\mathcal{D}} = \left\{ \pi_M : v^{\mathcal{D}}(\pi_M) > \Lambda_2^{\mathcal{D}} \right\} \subseteq \left\{ \pi_M : v^{\mathcal{D}}(\pi_M) > \Lambda_1^{\mathcal{D}} \right\} = \Pi_1^{\mathcal{D}}.$$

This proves part (i).

Step 2: The crossing point.

For every $\pi_M \in \Pi_2^{\mathcal{D}}$, the rule belongs to both consideration sets. Hence

$$\begin{aligned}\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M) &= \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_2^{\mathcal{D}}}{\kappa_2(\pi_M)} - \frac{v^{\mathcal{D}}(\pi_M) - \Lambda_1^{\mathcal{D}}}{\gamma\kappa_2(\pi_M)} \\ &= \frac{\gamma - 1}{\gamma\kappa_2(\pi_M)}\left[v^{\mathcal{D}}(\pi_M) - \bar{v}^{\mathcal{D}}\right],\end{aligned}$$

where

$$\bar{v}^{\mathcal{D}} := \frac{\gamma\Lambda_2^{\mathcal{D}} - \Lambda_1^{\mathcal{D}}}{\gamma - 1}.$$

The affine term in brackets has the unique zero $\bar{v}^{\mathcal{D}}$. Moreover,

$$\bar{v}^{\mathcal{D}} - \Lambda_2^{\mathcal{D}} = \frac{\Lambda_2^{\mathcal{D}} - \Lambda_1^{\mathcal{D}}}{\gamma - 1} > 0.$$

Thus, for every $\pi_M \in \Pi_2^{\mathcal{D}}$,

$$\operatorname{sgn}\Big(\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\Big) = \operatorname{sgn}\Big(v^{\mathcal{D}}(\pi_M) - \bar{v}^{\mathcal{D}}\Big).$$

Now consider $\pi_M \in \Pi_1^{\mathcal{D}} \setminus \Pi_2^{\mathcal{D}}$. By the definition of $\Pi_2^{\mathcal{D}}$, $v^{\mathcal{D}}(\pi_M) \leq \Lambda_2^{\mathcal{D}} < \bar{v}^{\mathcal{D}}$, while $\rho_2^{\mathcal{D}}(\pi_M) = 0 < \rho_1^{\mathcal{D}}(\pi_M)$. Therefore, the same sign characterization holds on $\Pi_1^{\mathcal{D}} \setminus \Pi_2^{\mathcal{D}}$. Hence, for every $\pi_M \in \Pi_1^{\mathcal{D}}$,

$$\operatorname{sgn}\Big(\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\Big) = \operatorname{sgn}\Big(v^{\mathcal{D}}(\pi_M) - \bar{v}^{\mathcal{D}}\Big).$$

If $\pi_M \notin \Pi_1^{\mathcal{D}}$, part (i) implies that $\pi_M \notin \Pi_2^{\mathcal{D}}$ as well, and therefore $\rho_2^{\mathcal{D}}(\pi_M) = \rho_1^{\mathcal{D}}(\pi_M) = 0$.

It remains to locate the crossing point. Let $\underline{v}_1^{\mathcal{D}} := \min_{\pi_M \in \Pi_1^{\mathcal{D}}} v^{\mathcal{D}}(\pi_M)$ and $\overline{v}^{\mathcal{D}} := v^{\mathcal{D}}(\pi_{M^*}^{\max})$. Every maximal-value rule belongs to both consideration sets, and $\overline{v}^{\mathcal{D}}$ is the largest value attained on $\Pi_1^{\mathcal{D}}$. Since both extraction distributions sum to one, $\sum_{\pi_M \in \Pi}\left[\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\right] = 0$.

Suppose first that $\bar{v}^{\mathcal{D}} \leq \underline{v}_1^{\mathcal{D}}$. The sign characterization then implies that every nonzero probability difference is nonnegative. Since $\Pi_1^{\mathcal{D}}$ contains rules attaining at least two distinct values, at least one of these differences is strictly positive. This contradicts the zero-sum identity above.

Similarly, suppose that $\bar{v}^{\mathcal{D}} \geq \overline{v}^{\mathcal{D}}$. Every nonzero probability difference is then nonpositive. Since $\Pi_1^{\mathcal{D}}$ contains at least two distinct value levels, at least one rule has value strictly below $\overline{v}^{\mathcal{D}}$, and its probability difference is strictly negative. This again contradicts the zero-sum identity. Consequently,

$$\min_{\pi_M \in \Pi_1^{\mathcal{D}}} v^{\mathcal{D}}(\pi_M) < \bar{v}^{\mathcal{D}} < v^{\mathcal{D}}(\pi_{M^*}^{\max}).$$

This proves part (ii).

Step 3: First-order stochastic dominance.

Fix $z \in \mathbb{R}$ such that $0 < F_1^{\mathcal{D}}(z) < 1$. Since every rule outside $\Pi_1^{\mathcal{D}}$ receives zero probability

under both types, we have

$$F_2^{\mathcal{D}}(z) - F_1^{\mathcal{D}}(z) = \sum_{\substack{\pi_M \in \Pi_1^{\mathcal{D}}: \\ v^{\mathcal{D}}(\pi_M) \leq z}} \left[\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\right].$$

Suppose first that $z < \bar{v}^{\mathcal{D}}$. For every $\pi_M \in \Pi_1^{\mathcal{D}}$ satisfying $v^{\mathcal{D}}(\pi_M) \leq z$, the sign characterization in part (ii) implies $\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M) < 0$. Moreover, because $F_1^{\mathcal{D}}(z) > 0$, there exists at least one $\pi_M \in \Pi_1^{\mathcal{D}}$ satisfying $v^{\mathcal{D}}(\pi_M) \leq z$. Therefore, $F_2^{\mathcal{D}}(z) - F_1^{\mathcal{D}}(z) < 0$.

Suppose instead that $z \geq \bar{v}^{\mathcal{D}}$. Since the probability differences sum to zero,

$$F_2^{\mathcal{D}}(z) - F_1^{\mathcal{D}}(z) = -\sum_{\substack{\pi_M \in \Pi_1^{\mathcal{D}}: \\ v^{\mathcal{D}}(\pi_M) > z}} \left[\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M)\right].$$

For every rule appearing in the sum, $v^{\mathcal{D}}(\pi_M) > z \geq \bar{v}^{\mathcal{D}}$, so the sign characterization in part (ii) implies $\rho_2^{\mathcal{D}}(\pi_M) - \rho_1^{\mathcal{D}}(\pi_M) > 0$. Moreover, because $F_1^{\mathcal{D}}(z) < 1$, there exists at least one $\pi_M \in \Pi_1^{\mathcal{D}}$ satisfying $v^{\mathcal{D}}(\pi_M) > z$. Hence, $F_2^{\mathcal{D}}(z) - F_1^{\mathcal{D}}(z) < 0$.

Thus, $F_2^{\mathcal{D}}(z) < F_1^{\mathcal{D}}(z)$ whenever $0 < F_1^{\mathcal{D}}(z) < 1$. By definition, $\rho_2^{\mathcal{D}} \succeq_{\text{FOSD}}^{v^{\mathcal{D}}} \rho_1^{\mathcal{D}}$. This proves part (iii).

A(v).c Proof of Corollary 2

Part (a). Since $\rho_1^{\mathcal{D}}$ is a probability distribution, there exists some $\widehat{\pi}_{\widehat{M}} \in \Pi_1^{\mathcal{D}}$ such that $\rho_1^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}}) > 0$. By the definition of the consideration set, $v^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}}) > \Lambda(\Pi_1^{\mathcal{D}})$. Since $\pi_{M^*}^{\max}$ maximizes value,

$$v^{\mathcal{D}}(\pi_{M^*}^{\max}) \geq v^{\mathcal{D}}(\widehat{\pi}_{\widehat{M}}) > \Lambda(\Pi_1^{\mathcal{D}}),$$

and therefore $\pi_{M^*}^{\max} \in \Pi_1^{\mathcal{D}}$.

By Proposition 4(ii), $\bar{v}^{\mathcal{D}} < v^{\mathcal{D}}(\pi_{M^*}^{\max})$. Applying the sign characterization in Proposition 4(ii) to $\pi_{M^*}^{\max}$ gives

$$\operatorname{sgn}\Big(\rho_2^{\mathcal{D}}(\pi_{M^*}^{\max}) - \rho_1^{\mathcal{D}}(\pi_{M^*}^{\max})\Big) = \operatorname{sgn}\Big(v^{\mathcal{D}}(\pi_{M^*}^{\max}) - \bar{v}^{\mathcal{D}}\Big) = 1.$$

Hence, $\rho_2^{\mathcal{D}}(\pi_{M^*}^{\max}) > \rho_1^{\mathcal{D}}(\pi_{M^*}^{\max})$.

Part (b). Suppose first that $\rho_1^{\mathcal{D}}(\pi^R) > 0$. Then $\pi^R \in \Pi_1^{\mathcal{D}}$. Since $v^{\mathcal{D}}(\pi^R) < \bar{v}^{\mathcal{D}}$, the sign characterization in Proposition 4(ii) implies

$$\operatorname{sgn}\Big(\rho_2^{\mathcal{D}}(\pi^R) - \rho_1^{\mathcal{D}}(\pi^R)\Big) = \operatorname{sgn}\Big(v^{\mathcal{D}}(\pi^R) - \bar{v}^{\mathcal{D}}\Big) = -1.$$

Therefore, $\rho_2^{\mathcal{D}}(\pi^R) < \rho_1^{\mathcal{D}}(\pi^R)$.

Suppose instead that $\rho_1^{\mathcal{D}}(\pi^R) = 0$. By the definition of the consideration set, $\pi^R \notin \Pi_1^{\mathcal{D}}$. Proposition 4(i) gives $\Pi_2^{\mathcal{D}} \subseteq \Pi_1^{\mathcal{D}}$, and hence $\pi^R \notin \Pi_2^{\mathcal{D}}$. Consequently, $\rho_2^{\mathcal{D}}(\pi^R) = \rho_1^{\mathcal{D}}(\pi^R) = 0$. Combining the two cases yields $\rho_2^{\mathcal{D}}(\pi^R) \leq \rho_1^{\mathcal{D}}(\pi^R)$, with strict inequality whenever $\rho_1^{\mathcal{D}}(\pi^R) > 0$.

B ADDITIONAL FIGURES AND TABLES

B(i) DATA DESIGN

Treatment	Alone	Or	Inhibit	Either	Joint
Two lights					
	HAHG	HALG	LAHG	LAHG	HALG
T1	87.50 / 71.88	87.10 / 83.87	82.14 / 67.86	82.76 / 68.97	89.66 / 86.21
	HALG	LAHG	HAHG	HALG	HAHG
T2	86.21 / 82.76	80.00 / 70.00	93.55 / 77.42	86.67 / 83.33	88.89 / 70.37
	LAHG	HAHG	HALG	HAHG	LAHG
T3	80.00 / 70.00	87.50 / 71.88	90.32 / 87.10	90.00 / 73.33	83.33 / 73.33
Three lights					
	HALG	HAHG	HAHG	HALG	HALG
T1	86.21 / 82.76 / 86.21	87.50 / 71.88 / 81.25	93.55 / 77.42 / 87.10	86.67 / 83.33 / 86.67	89.66 / 86.21 / 89.66
	HAHG	HALG	HALG	HAHG	HAHG
T2	87.50 / 75.00 / 81.25	87.10 / 83.87 / 87.10	90.32 / 87.10 / 90.32	90.00 / 73.33 / 83.33	88.89 / 70.37 / 81.48

Notes: Each cell reports the accuracy-gap condition followed by rule accuracies in percent. Entries are ordered as *DGP / best non-DGP rule using at most two lights / best non-DGP rule among all rules*. The third number is shown only when the two non-DGP accuracies differ. The restricted set contains all 16 deterministic mappings in the two-light experiment and the 38 mappings using at most two lights in the three-light experiment; the unrestricted three-light set contains all 256 deterministic mappings.

Table B1: Value structure by accuracy-gap condition and machine

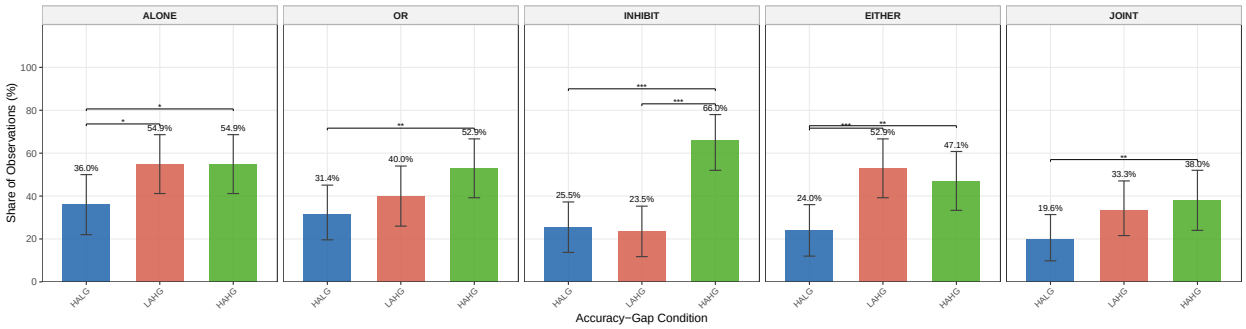


Figure B1: Highest-value rule extraction by machine and accuracy-gap condition in the two-light experiment. Bars report the shares of subject–dataset observations classified by exact binomial tests as extracting the highest-value rule. Error bars show 95% subject-level bootstrap confidence intervals.

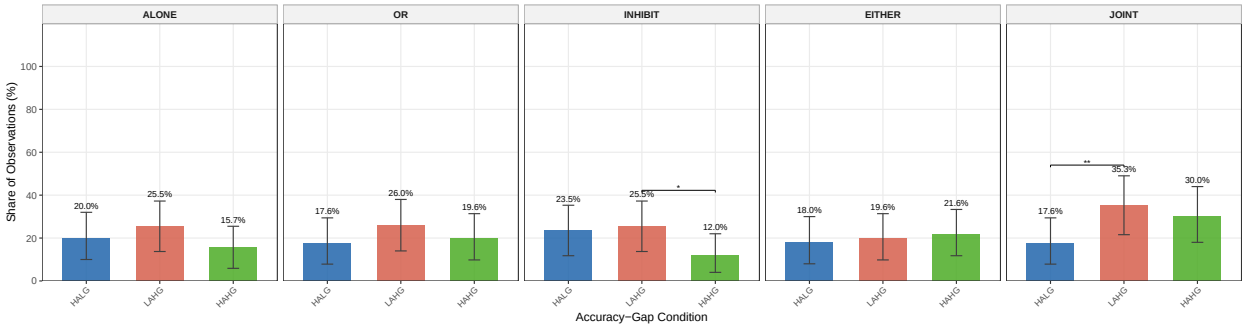


Figure B2: Randomization by machine and accuracy-gap condition in the two-light experiment. Bars report the shares of subject–dataset observations classified as consistent with random choice. Error bars show 95% subject-level bootstrap confidence intervals.

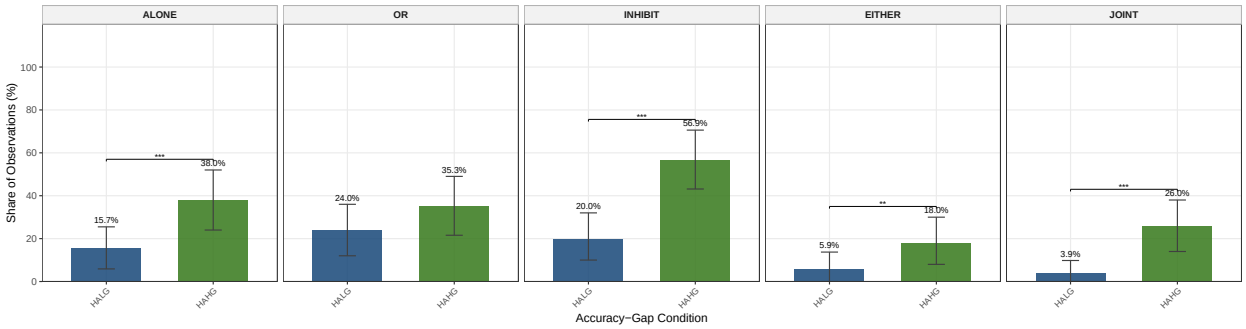


Figure B3: Highest-value rule extraction by machine and accuracy-gap condition in the three-light experiment. Bars report the shares of subject–dataset observations classified by exact binomial tests as extracting the highest-value rule. Error bars show 95% subject-level bootstrap confidence intervals.

Accuracy-Gap Condition	Two-Light Experiment				Three-Light Experiment			
	Average value (%)	Failure frequency (%)	Conditional severity $\bar{\Delta}$ (p.p.)	Expected loss $\mathcal{L}$ (p.p.)	Average value (%)	Failure frequency (%)	Conditional severity $\bar{\Delta}$ (p.p.)	Expected loss $\mathcal{L}$ (p.p.)
HALG	78.7	64.9	14.3	9.3	78.9	80.6	11.3	9.1
LAHG	72.2	45.5	20.8	9.5				
HAHG	79.0	40.2	26.1	10.5	73.0	68.2	24.2	16.5

Notes: Average value is the expected value of the extracted rule, computed using the maximum-likelihood estimates of the population rule shares. Failure frequency is the probability of extracting a rule whose value is strictly below that of the DGP-implied highest-value rule; value-equivalent rules in the three-light HALG are therefore not counted as extraction failures. Conditional severity, $\bar{\Delta}$, is the average value loss, in percentage points, conditional on an extraction failure. Expected extraction loss is $\mathcal{L}$ = Failure frequency $\cdot \bar{\Delta}$. Random-choice rules are assigned a value of 0.5. Estimates are aggregated using the number of subjects as weights.

Table B2: The Frequency–Severity Decomposition Based on Maximum-Likelihood Rule Shares

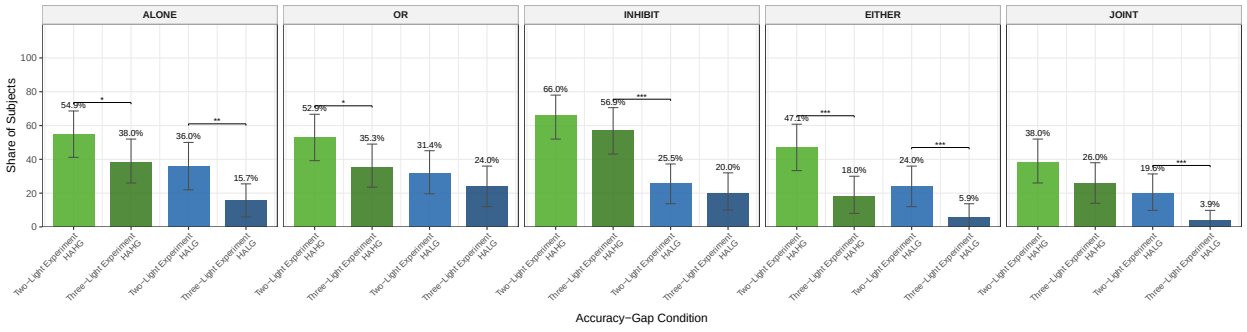
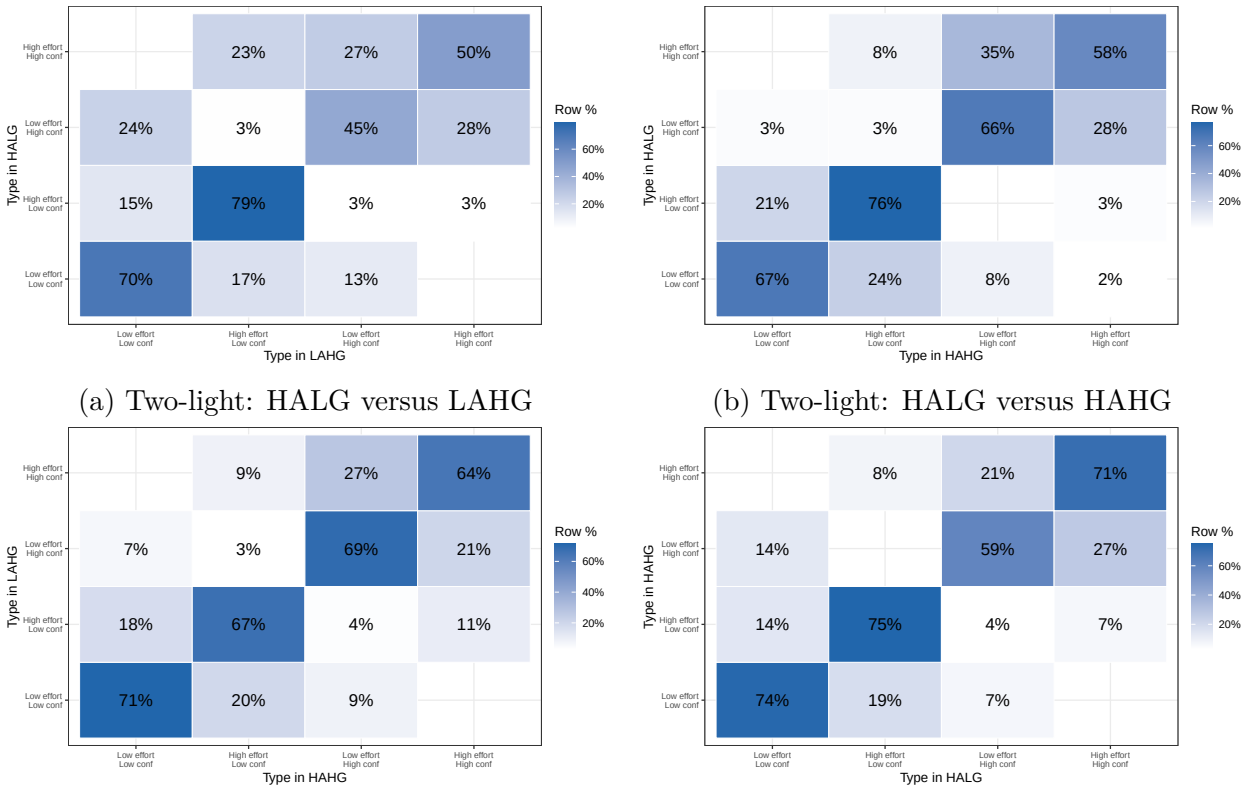


Figure B4: Highest-value rule extraction by machine, experiment, and accuracy-gap condition. Bars report the shares of subject–dataset observations classified by exact binomial tests as extracting the highest-value rule in the two- and three-light experiments. Error bars show 95% subject-level bootstrap confidence intervals.

Experiment	Two-Light Experiment			Three-Light Experiment		
Outcome	All Rules	True Extracted	Randomizing	All Rules	True Extracted	Randomizing
	(1)	(2)	(3)	(4)	(5)	(6)
Condition: LAHG	-0.841***	14.6***	6.44**			
	(0.177)	(3.85)	(3.08)			
Condition: HAHG	1.00**	23.6***	-0.297	2.85***	21.5***	-0.584
	(0.496)	(4.05)	(3.26)	(0.481)	(3.95)	(3.53)
Distance (p.p.)	-0.645***			-0.143***		
	(0.061)			(0.049)		
HELC	3.92**	-2.43	-3.41	1.29	16.4*	-0.119
	(1.94)	(7.60)	(6.65)	(1.48)	(9.59)	(11.0)
LEHC	3.07	21.9***	-10.3	1.52	37.2***	0.214
	(2.27)	(7.07)	(6.64)	(1.74)	(10.7)	(11.1)
HEHC	6.49***	28.9***	-12.1	3.55**	23.1*	7.86
	(2.47)	(9.47)	(8.31)	(1.53)	(13.3)	(11.6)
Distance × HELC	-0.197**			-0.067		
	(0.082)			(0.071)		
Distance × LEHC	-0.050			-0.050		
	(0.099)			(0.086)		
Distance × HEHC	-0.259**			-0.240***		
	(0.106)			(0.067)		
Subjects	152	152	152	101	101	101
Observations	4,761	760	760	5,095	505	505
R^2	0.128	0.370	0.367	0.020	0.328	0.432
Within R^2	0.127	0.073	0.013	0.019	0.102	0.002
Subject fixed effects	✓	✓	✓	✓	✓	✓

Notes: Each model is estimated separately for the indicated experiment and includes subject fixed effects. Standard errors in parentheses are heteroskedasticity-robust and clustered by subject. The *All Rules* columns use posterior assignment probabilities, so each subject enters once for every candidate rule in each dataset; these probabilities sum to one within each subject–dataset pair. These specifications include interactions between Distance and subject type. The other columns use binary assignments from the prespecified test for extraction of the highest-value rule and the exact random-choice test. The omitted subject type is LELC. HELC, LEHC, and HEHC denote high effort/low confidence, low effort/high confidence, and high effort/high confidence, respectively. HALG is the omitted condition in both experiments.
$^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table B3: Rule Extraction by Accuracy and Subjective Task Difficulty: Subject Fixed Effects



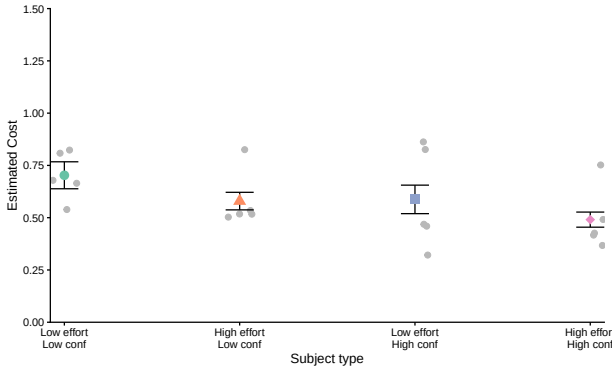
(a) Two-light: HALG versus LAHG

(b) Two-light: HALG versus HAHG

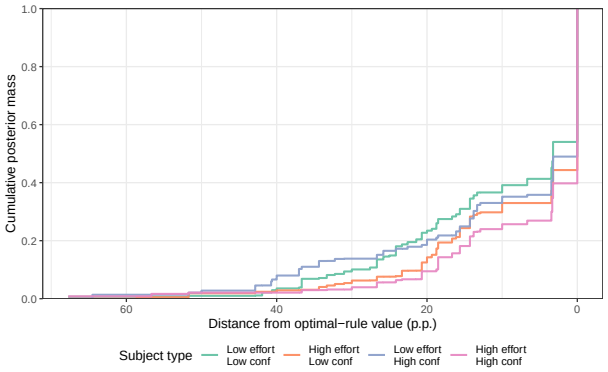
(c) Two-light: LAHG versus HAHG

(d) Three-light: HALG versus HAHG

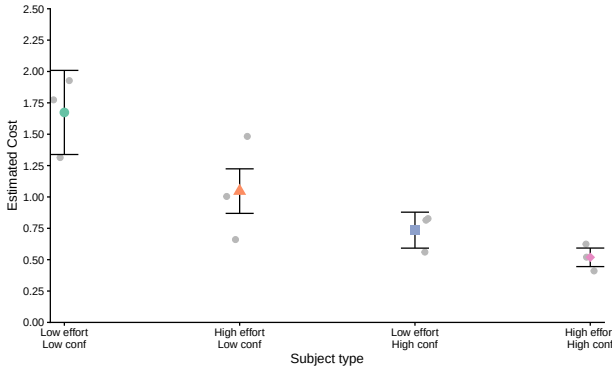
Figure B5: Stability of subjective-difficulty classifications across accuracy-gap conditions. Each panel cross-tabulates the effort–confidence type assigned in the row condition against the type assigned in the column condition. Cells report row percentages; diagonal cells indicate stable classifications, and blank cells indicate zero observations.



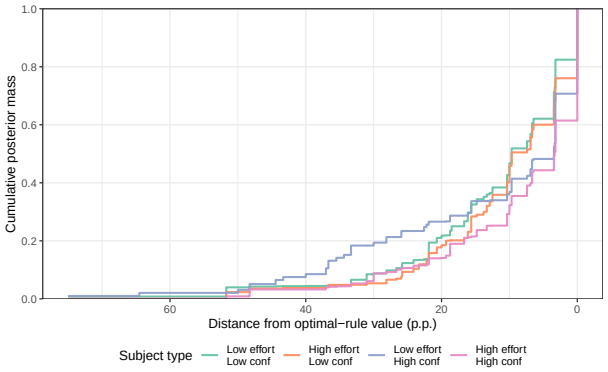
(a) Two-light: estimated costs



(b) Two-light: rule-value distribution



(c) Three-light: estimated costs, restricted rule set



(d) Three-light: rule-value distribution

Figure B6: Estimated costs and rule-value distributions by experiment and subject type. Panels (a) and (c) report structurally estimated costs recovered from posterior rule assignments; gray points show rule-specific estimates, colored markers show averages by subject type, and error bars show 95% subject-level bootstrap confidence intervals. Panels (b) and (d) report posterior-weighted cumulative distributions of extracted rules by value distance from the highest-value rule. Three-light cost estimates exclude Either and Joint.

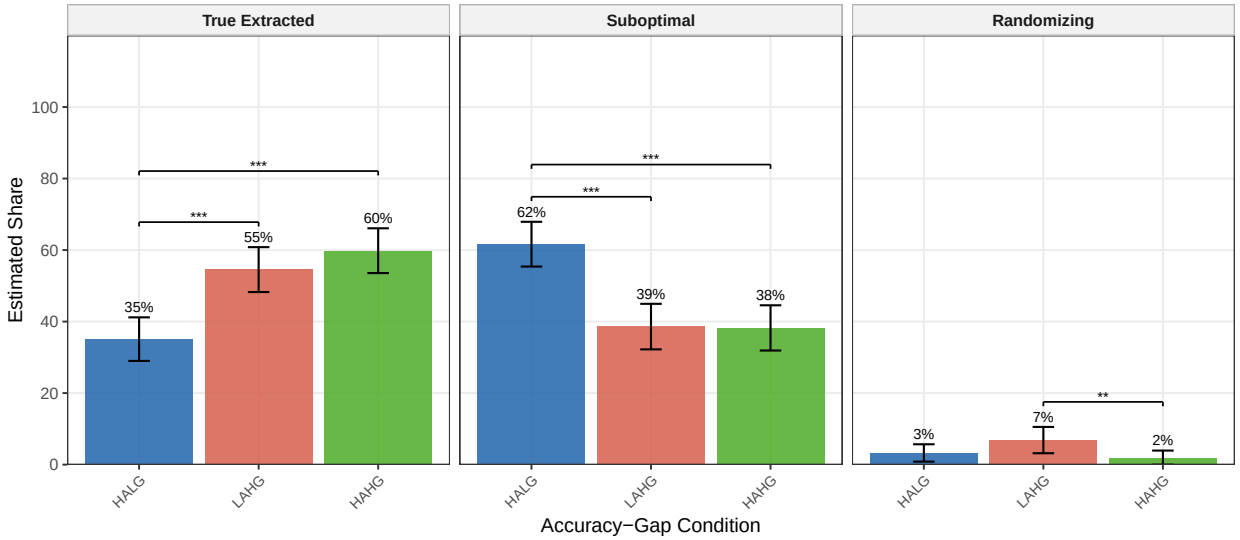


Figure B7: Maximum-likelihood rule shares across accuracy-gap conditions in the two-light experiment. Bars report estimated population shares assigned to the highest-value rule, alternative deterministic rules, and randomization, pooling the five machines. Error bars show 95% model-based confidence intervals; brackets report two-sided Wald tests.

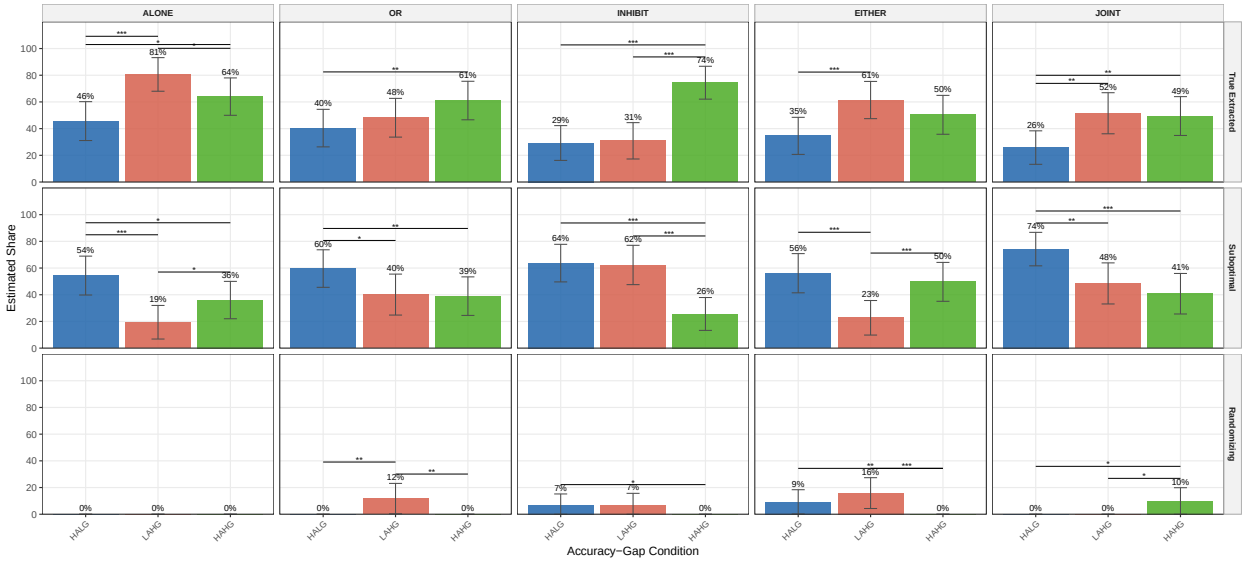


Figure B8: Maximum-likelihood rule shares by machine and accuracy-gap condition in the two-light experiment. Bars report estimated population shares assigned to the highest-value rule, alternative deterministic rules, and randomization separately for each machine. Error bars show 95% model-based confidence intervals; brackets report two-sided Wald tests.

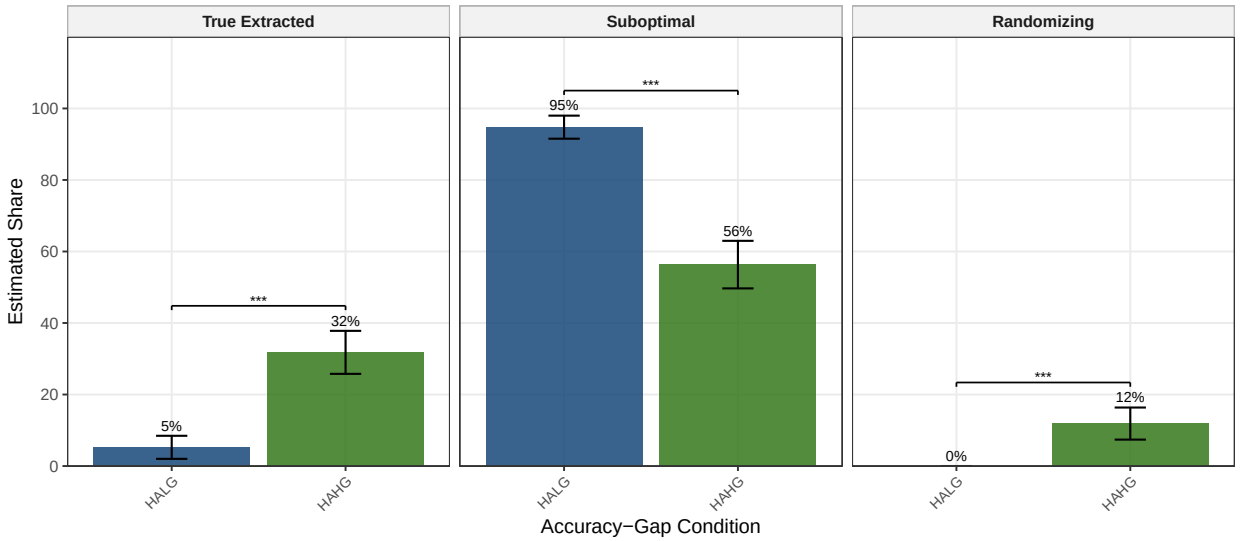


Figure B9: Maximum-likelihood rule shares across accuracy-gap conditions in the three-light experiment. Bars report estimated population shares assigned to the highest-value rule, alternative deterministic rules, and randomization, pooling the five machines. Error bars show 95% model-based confidence intervals; brackets report two-sided Wald tests.

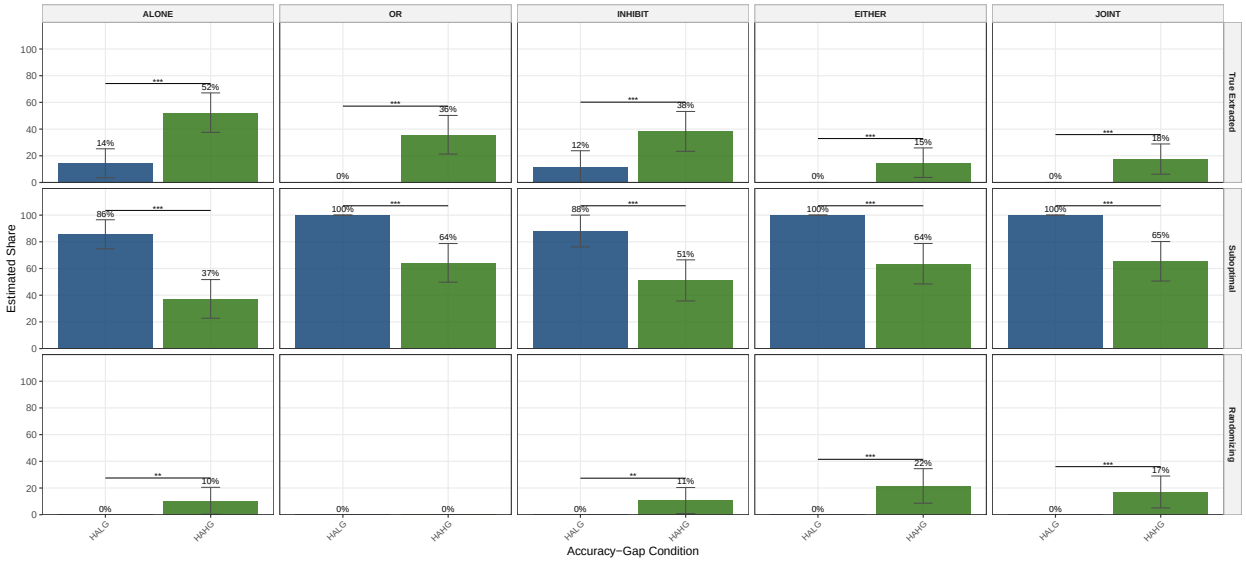


Figure B10: Maximum-likelihood rule shares by machine and accuracy-gap condition in the three-light experiment. Bars report estimated population shares assigned to the highest-value rule, alternative deterministic rules, and randomization separately for each machine. Error bars show 95% model-based confidence intervals; brackets report two-sided Wald tests.

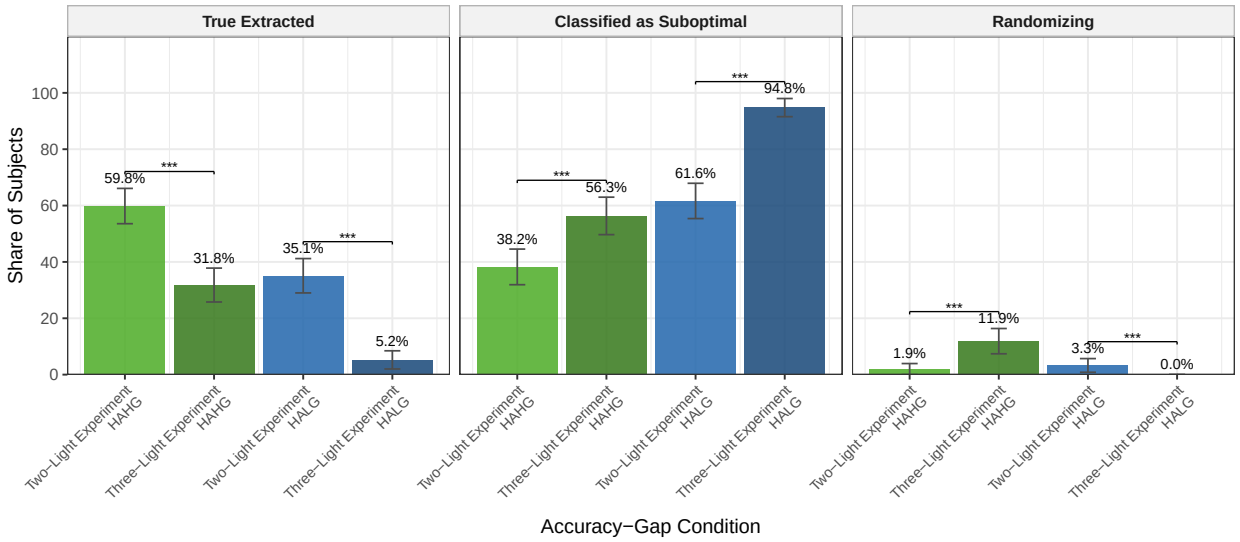


Figure B11: Maximum-likelihood rule shares by experiment and accuracy-gap condition. Bars report estimated population shares assigned to the highest-value rule, alternative deterministic rules, and randomization in the two- and three-light experiments. Error bars show 95% model-based confidence intervals; brackets report two-sided Wald tests.

B(iv) RE-ANALYSIS OF KENDALL AND OPREA (2024)

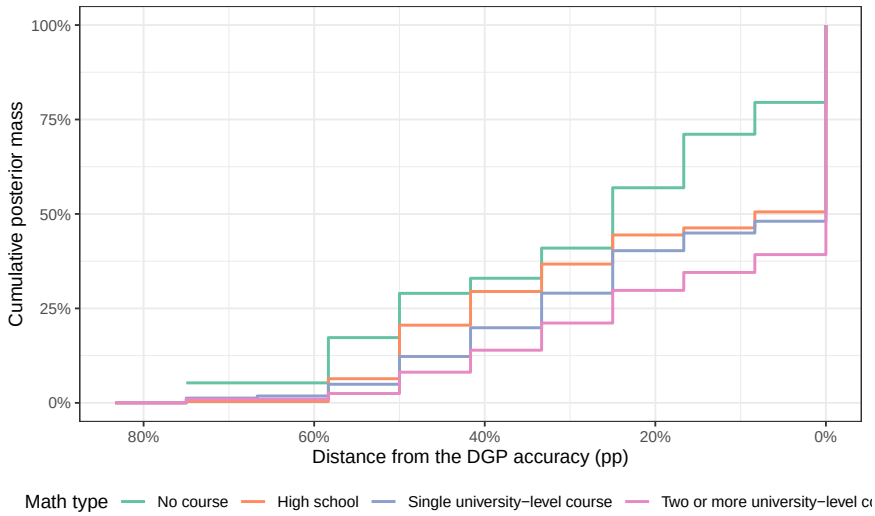


Figure B12: Rule-value distributions by mathematical education in the Kendall and Oprea (2024) experiment. Curves report posterior-weighted cumulative distributions over deterministic automata by value distance from the DGP, measured as the distance in learning-sample accuracy. Datasets from the *High Gap* and *Low Gap* conditions are pooled.

C DGPs IN KENDALL AND OPREA (2024)

Kendall and Oprea (2024) study an agent who observes a dataset consisting of (i) a binary "input" variable $i_t \in I = \{a, b\}$ observed at dates $t = 1, \ldots, T$ and (ii) a binary "output" variable $y_t \in Y = \{x, y\}$ observed at dates $t = 0, \ldots, T$. The data are generated by a data–generating process (DGP) that produces an initial output y_0 and, for each $t \geq 1$, an output y_t as a function of past inputs and outputs.[58] Since each DGP is deterministic, it coincides with the highest-value prediction rule; we therefore use the terms DGP and highest-value rule interchangeably in this section. Consistent with our theoretical framework, Kendall and Oprea (2024) focus on DGPs that (i) have discrete inputs and outputs, (ii) are deterministic, and (iii) exhibit a (weakly) one–directional causal link from histories to outputs, with a history at time t given by

$$h_t = (y_0,\, i_1,\, y_1,\, i_2,\, \ldots,\, y_{t-1},\, i_t),$$

[58] Formally, the DGP is a finite automaton described by the four–tuple (S, s^0, f, τ), where S is a finite set of states, $s^0 \in S$ is the initial state, $f : S \to Y$ is an output function specifying the output in each state, and $\tau : S \times I \to S$ is a transition function mapping a current state and input $i \in I$ to the subsequent state.

and the set of possible histories being

$$\mathcal{H}_t = \prod_{j=1}^{t} (Y_{j-1} \times I_j) .$$

Define $M_I \subseteq \{1, \ldots, t\}$ and $M_Y \subseteq \{0, \ldots, t-1\}$. A rule π_{M_I, M_Y} indicates a mapping that is measurable with respect to relevant lags for past inputs and outputs. In other words it defines the memory of the strategy.

In these diagrams, (i) each circle represents a state, $s \in S$, (ii) each arc a transition, (iii) letters next to arcs show inputs $i \in I$, and (iv) letters inside circles represent outputs, $y \in Y$, in each state. A freestanding arrow pointing to one of the states indicates the initial state, s^0. For instance, automaton I2 (the 2-state "Instruct" DGP) specifies that (i) output x appears at date 0 and (ii) afterwards output x always appears in the period following input b while y always follows input a.

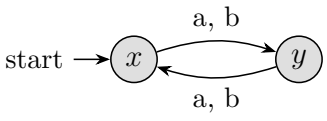
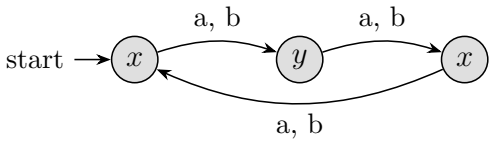
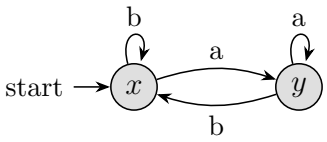
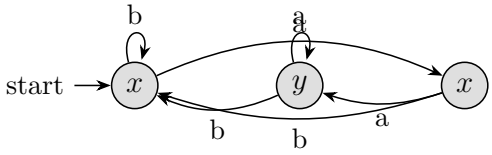
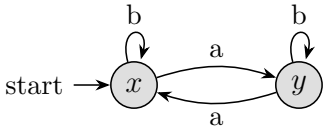
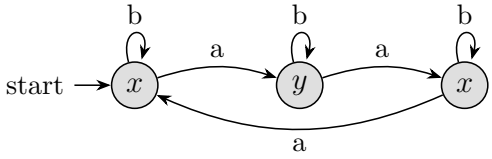
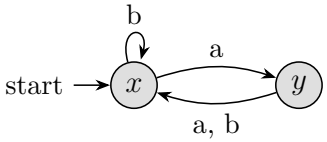
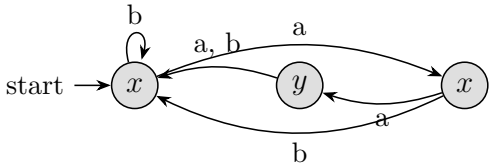
DGP	**2-State DGP**	**3-State Extension**
Autonomous *A fixed pattern, independent of inputs*	 A2	 A3
Instruct *A direct mapping of previous input to output*	 I2	 I3
Switch *A direction to move back or forth over a sequence of outputs*	 S2	 S3
Hybrid *One input instructs, the other prescribes a switch*	 H2	 H3

Table C1: Data-generating process (DGP) classes and their 2- and 3-state counterparts.

C(i) COMPLEXITY MEASURES

Sensitivity. Sensitivity measures how strongly the output depends on the history of inputs. It counts, across possible input histories, the one-period changes to an input that would change the resulting output, with larger values indicating that more features of the input history matter for prediction.

Mutual information. Mutual information measures the statistical dependence between the output and the information available to predict it. The three variants in Figure 7 use, respectively, the current input (*mutual_1*), the recent input history (*mutual_max*), and the previous output (*mutual_out*); in each case, a larger value means that the conditioning information reduces more uncertainty about the output.

Entropy. Entropy is the Shannon entropy of the output generated by the automaton. It measures the variability, or surprise, in the output sequence: entropy is low when one output occurs almost always and high when the two outputs occur with similar frequencies.

States and transitions. State complexity is the number of states in the automaton representing the DGP, while transition complexity is the number of arcs needed to describe how inputs move the automaton between states. These measures capture the implementation burden of storing the current state and applying the DGP's transition rule.

Machine complexity. Machine complexity, labeled *computational* in Figure 7, is the minimum number of two-input logic gates required to implement the automaton physically. It therefore measures the hardware resources needed to execute the DGP.

Partition complexity. Partition complexity is the number of elements in the coarsest partition of data subhistories that permits correct prediction of the DGP's output. Equivalently, it counts the distinct patterns that a decision maker must distinguish: for example, I2 requires two patterns, whereas S2 requires four because prediction depends on both the input and the previous output.

Sparsity complexity. Sparsity complexity is the minimum number of input and output elements in the data history to which a decision maker must attend in order to predict correctly. Thus, I2 requires attention only to the current input, whereas S2 requires attention to both the current input and the previous output. Unlike partition complexity, which counts distinct patterns, sparsity complexity counts the pieces of information used to construct them.

D IDENTIFICATION AND ESTIMATION

D(i) BINOMIAL TEST

For each subject–dataset observation let M_{id} denote the number of predictions made by subject i in dataset d that coincide with the highest-value rule. Under the null hypothesis of independent uniform random choice, M_{id} follows a binomial distribution with success probability one half. We classify the highest-value rule as *extracted* when the two-sided exact binomial p-value is weakly below 0.01 and M_{id} exceeds one half of the predictions. This criterion requires at least 11 matches out of 12 in the two-light experiment and at least 14 matches out of 16 in the three-light experiment.

For each subject–dataset observation that does not meet the highest-value rule extraction criterion, we define the test statistic as the maximum number of matches between the subject's predictions and any deterministic rule other than the highest-value rule. Because the same prediction sequence is compared with many overlapping rules and the best-fitting rule is selected ex post, this maximum does not follow a standard binomial distribution. Under the null hypothesis that predictions are independent Bernoulli(1/2) draws, we therefore compute its exact distribution using the realized frequencies of the light configurations and conditional on the highest-value rule not having satisfied the extraction criterion. The reported p-value is the conditional probability of obtaining a maximum agreement at least as large as that observed. By constructing the null distribution for the maximum directly, the test accounts for the search over all alternative deterministic rules: 15 in the two-light experiment and 255 in the three-light experiment.

The reported reduced-form shares are averages of these binary classifications across subject–dataset observations. Confidence intervals and tests are computed using a bootstrap at the subject level, separately from the bootstrap used for the MLE/posterior estimates. In each of 5,000 replications, subjects are resampled with replacement within each treatment. Confidence intervals are the 2.5th and 97.5th percentiles of the bootstrap distribution, and pairwise comparisons use two-sided centered-bootstrap p-values.

D(ii) MAXIMUM LIKELIHOOD ESTIMATION

We infer the prediction rules used by participants with a finite-mixture model, following the strategy-frequency approach developed in the experimental repeated-games literature and its adaptation to model extraction (Dal Bó and Fréchette, 2011; Fréchette et al., 2024). A prediction rule is a complete mapping from the observable information in a task to a binary prediction. The rule used by a participant is latent: the researcher observes only the participant's sequence of predictions and estimates both the population frequency of each candidate rule and the probability that each rule generated a given participant's choices.

D(ii).a Likelihood

Let i index participants, d datasets, and $t = 1, \ldots, T_d$ prediction trials. The observed prediction is $a_{idt} \in A = \{0, 1\}$. Let Π_d denote the set of candidate rules for dataset d. For a deterministic rule $\pi_M \in \Pi_d$, let $a_{dt}^{\pi_M} \in A$ denote the prediction prescribed for the configuration presented in trial t. We allow participants to depart from their rule, due to implementation mistakes, with probability q_d. Conditional on extracting rule π_M, these deviations are independent across trials and participants. The likelihood of participant i's prediction sequence is therefore

$$p_{id}(\pi_M) = \prod_{t=1}^{T_d} (1 - q_d)^{\mathbb{1}\{a_{idt} = a_{dt}^{\pi_M}\}} q_d^{\mathbb{1}\{a_{idt} \neq a_{dt}^{\pi_M}\}}. \tag{9}$$

The error probability is common across deterministic rules within a dataset. It gives every observed prediction sequence positive probability and distinguishes occasional deviations from the use of a different rule. The candidate set Π_d contains all deterministic mappings from the observed configurations to predictions, together with stochastic rules that do not condition on the lights. For a stochastic rule, equation (9) is replaced by the corresponding Bernoulli likelihood. In particular, we allow both uniform randomization and a rule that predicts the sound on with a calibrated probability q, π^q.

Let $\rho^{\mathcal{D}_d}(\pi_M) \geq 0$ denote the population frequency with which rule $\pi_M \in \Pi_d$ is extracted in decision problem $\mathcal{D}_d$, with

$$\sum_{\pi_M \in \Pi_d} \rho^{\mathcal{D}_d}(\pi_M) = 1.$$

Because the rule extracted by participant i is unobserved, we integrate over the candidate rules. The log-likelihood is

$$\mathcal{L} = \sum_d \sum_i \log \left[\sum_{\pi_M \in \Pi_d} \rho^{\mathcal{D}_d}(\pi_M) p_{id}(\pi_M) \right]. \tag{10}$$

Thus, $\rho^{\mathcal{D}_d}(\pi_M)$ is the estimated fraction of participants who extract rule π_M in decision problem $\mathcal{D}_d$; it is not the fraction of individual predictions that happen to coincide with that rule. The specification allows a participant to extract different rules in different datasets but holds the latent rule fixed across the repeated prediction trials within a dataset.

We maximize equation (10) using the expectation-maximization algorithm (Dvorak, 2022). Moreover, to avoid overfitting, we impose a baseline set containing every distinct deterministic rule that attains the highest-value in at least one of the experimental treatments. This set contains six rules in the two-light experiment—the red- and blue-Alone rules, together with Or, Inhibit, Either, and Joint—and five rules in the three-light experiment, where the Alone rule always uses the red light. For each dataset, we then add each remaining

candidate rule to the baseline one at a time. The candidate pool consists of all remaining deterministic mappings, uniform randomization, and a stochastic rule with a configuration-independent propensity to predict one. Let

$$\Delta\,\mathrm{BIC}(\pi_M) \equiv \mathrm{BIC}\Big(\Pi_d^0 \cup \{\pi_M\}\Big) - \mathrm{BIC}\Big(\Pi_d^0\Big),$$

where Π_d^0 denotes the baseline set. A candidate passes the initial screen only if $\Delta\,\mathrm{BIC}(\pi_M) < -2$, so that adding it lowers BIC by more than two points. Among the candidates that pass this screen, we use forward selection: at each step, we add the rule that produces the lowest BIC, provided that it further improves on the current specification, and stop when no remaining candidate lowers BIC. We compare the resulting specification with the unrestricted model, which contains all 16 deterministic rules and the two stochastic rules in the two-light experiment, and all 256 deterministic rules and the two stochastic rules in the three-light experiment. We retain whichever specification has the lower BIC. Finally, we remove rules assigned an estimated population share of exactly zero. For each remaining rule, we re-estimate the model after excluding that rule and test whether its exclusion significantly worsens model fit. Rules whose exclusion does not reduce fit significantly at the 5 percent level are removed. We then re-estimate the model using the resulting rule set. The procedure starts from rules motivated by the experimental design, then uses the data to select additional rules and prune rules that are not supported.

D(ii).b Posterior assignment and inference

Let $\widehat{\Pi}^{\mathcal{D}_d} \subseteq \Pi_d$ denote the final rule set retained by the model-selection procedure for decision problem $\mathcal{D}_d$. After estimating the mixture model, we use Bayes' rule to compute the posterior probability that participant i extracted rule $\pi_M \in \widehat{\Pi}^{\mathcal{D}_d}$:

$$\widehat{\tau}_{id}(\pi_M) = \frac{\widehat{\rho}^{\mathcal{D}_d}(\pi_M)\,p_{id}(\pi_M)}{\displaystyle\sum_{\pi'_{M'} \in \widehat{\Pi}^{\mathcal{D}_d}} \widehat{\rho}^{\mathcal{D}_d}(\pi'_{M'})\,p_{id}(\pi'_{M'})}, \qquad \sum_{\pi_M \in \widehat{\Pi}^{\mathcal{D}_d}} \widehat{\tau}_{id}(\pi_M) = 1. \tag{11}$$

The posterior combines two pieces of information: the estimated population frequency of rule π_M in decision problem $\mathcal{D}_d$ and the likelihood of participant i's prediction sequence conditional on that rule.

Maximum-a-posteriori (MAP) assignment. The MAP rule is the rule receiving the largest posterior probability:

$$\widehat{\pi}_{id}^{\mathrm{MAP}} = \arg \max_{\pi_M \in \widehat{\Pi}^{\mathcal{D}_d}} \widehat{\tau}_{id}(\pi_M).$$

The MAP classification assigns each participant–dataset observation to a single category: *True Extracted* if $\widehat{\pi}_{id}^{\mathrm{MAP}} = \pi_{M^*}^{\max}$, *Alternative Rules* if it is another deterministic rule, and *Randomizing* if it is a stochastic rule that does not condition on the light configuration. Assignments are based on the unrounded posterior probabilities.

Posterior-weighted assignment. The main structural analysis retains uncertainty when several rules explain a participant's predictions similarly. Let

$$\mathcal{C}_d = \left\{C_d^{\max}, C_d^{\mathrm{alt}}, C_d^{\mathrm{rand}}\right\}$$

denote the partition of the rule set into the highest-value rule, alternative deterministic rules, and stochastic non-conditioning rules. For any category $C \in \mathcal{C}_d$, define

$$\widehat{P}_{id}(C) = \sum_{\pi_M \in C \cap \widehat{\Pi}^{\mathcal{D}_d}} \widehat{\tau}_{id}(\pi_M), \qquad \sum_{C \in \mathcal{C}_d} \widehat{P}_{id}(C) = 1. \tag{12}$$

Posterior-weighted category shares are sample averages of equation (12). Thus, rather than assigning a participant–dataset observation entirely to its MAP rule, this approach allows it to contribute fractionally to the *True Extracted*, *Alternative Rules*, and *Randomizing* categories.

Standard Errors. Inference uses a subject-level bootstrap. Within each treatment, subjects are sampled with replacement and all their datasets are retained, preserving the dependence generated by the within-subject design. In each of 2,000 replications, we recompute the posterior-weighted shares and contrasts while holding the selected mixture model and posterior assignments fixed. The resulting procedure captures sampling variation across subjects conditional on the estimated model. Confidence intervals are percentile intervals, and pairwise comparisons use two-sided centered-bootstrap p-values. Population mixture-share figures instead use the covariance matrix of the maximum-likelihood estimates and two-sided Wald tests, while the regression tables report heteroskedasticity-robust standard errors clustered by subject.

D(iii) CLASSIFICATION OF TYPES

To study how rule extraction varies with costs, we construct subject types using measures of effort and confidence that do not depend on the rule assigned to the subject. Adapting the approach of Agranov et al. (2025), we interpret the combination of these measures as a proxy for the subjective complexity of the task.

Let $E \in \{2\text{L}, 3\text{L}\}$ denote the experiment and let $\mathcal{C}$ denote an accuracy-gap condition available in that experiment:

$$\mathcal{C} \in \mathcal{A}_E, \qquad \mathcal{A}_{2\text{L}} = \{\text{HALG}, \text{LAHG}, \text{HAHG}\}, \qquad \mathcal{A}_{3\text{L}} = \{\text{HALG}, \text{HAHG}\}.$$

For each subject i and experiment–condition cell $(E, \mathcal{C})$, we average three measures across the corresponding machine tasks. The first is learning-stage completion time, denoted by $e_i^{E,\mathcal{C}}$, which serves as our measure of effort. The second is the number of predictions the subject believes to have answered correctly, denoted by $a_i^{E,\mathcal{C}}$, which measures self-assessed accuracy. The third is the subject's reported confidence in that assessment, denoted by $c_i^{E,\mathcal{C}}$.

We compare each measure with its mean across subjects in the same experiment and accuracy-gap condition:

$$\bar{e}^{E,\mathcal{C}} \equiv \frac{1}{N_{E,\mathcal{C}}} \sum_i e_i^{E,\mathcal{C}}, \qquad \bar{a}^{E,\mathcal{C}} \equiv \frac{1}{N_{E,\mathcal{C}}} \sum_i a_i^{E,\mathcal{C}}, \qquad \bar{c}^{E,\mathcal{C}} \equiv \frac{1}{N_{E,\mathcal{C}}} \sum_i c_i^{E,\mathcal{C}},$$

where $N_{E,\mathcal{C}}$ denotes the number of subjects observed in cell $(E, \mathcal{C})$. We define the high-effort indicator as

$$H_{i,e}^{E,\mathcal{C}} \equiv \mathbb{1}\left\{e_i^{E,\mathcal{C}} > \bar{e}^{E,\mathcal{C}}\right\},$$

and the high-confidence indicator as

$$H_{i,c}^{E,\mathcal{C}} \equiv \mathbb{1}\left\{a_i^{E,\mathcal{C}} > \bar{a}^{E,\mathcal{C}} \text{ and } c_i^{E,\mathcal{C}} > \bar{c}^{E,\mathcal{C}}\right\}.$$

Thus, effort is classified as high when a subject's average completion time exceeds the corresponding cell mean. Confidence is classified as high only when both the subject's self-assessed accuracy and confidence in that assessment exceed their respective cell means. Ties are classified as low.

Combining these indicators yields a 2×2 classification of subjects' perceptions of task complexity. For each $(i, E, \mathcal{C})$, define

$$T_i^{E,\mathcal{C}} \in \{\text{EASY}, \text{DOABLE}, \text{HARD}, \text{TOO HARD}\}$$

according to

$$T_i^{E,\mathcal{C}} = \begin{cases} \text{EASY} & \text{if } H_{i,e}^{E,\mathcal{C}} = 0 \text{ and } H_{i,c}^{E,\mathcal{C}} = 1, \\ \text{DOABLE} & \text{if } H_{i,e}^{E,\mathcal{C}} = 1 \text{ and } H_{i,c}^{E,\mathcal{C}} = 1, \\ \text{HARD} & \text{if } H_{i,e}^{E,\mathcal{C}} = 1 \text{ and } H_{i,c}^{E,\mathcal{C}} = 0, \\ \text{TOO HARD} & \text{if } H_{i,e}^{E,\mathcal{C}} = 0 \text{ and } H_{i,c}^{E,\mathcal{C}} = 0. \end{cases}$$

Each subject is therefore assigned exactly one perception type in each experiment–accuracy-gap condition. Table D1 reports the distribution.

Experiment	Condition	Easy	Doable	Hard	Too Hard	Subjects
Two lights	HALG	29 (19.1%)	26 (17.1%)	34 (22.4%)	63 (41.4%)	152
	LAHG	29 (19.1%)	22 (14.5%)	45 (29.6%)	56 (36.8%)	152
	HAHG	33 (21.7%)	25 (16.4%)	44 (28.9%)	50 (32.9%)	152
Three lights	HALG	21 (20.8%)	25 (24.8%)	28 (27.7%)	27 (26.7%)	101
	HAHG	22 (21.8%)	24 (23.8%)	28 (27.7%)	27 (26.7%)	101

Notes: Cells report the number of subjects assigned to each subjective-complexity type, with the percentage of subjects in the corresponding experiment–accuracy-gap cell in parentheses. Because types are condition-specific, the same subject may be assigned to different types across accuracy-gap conditions.

Table D1: Distribution of Subject Types across Accuracy-Gap Conditions

We use these categories as proxies for heterogeneity in costs, treating *Easy* and *Doable* as lower-subjective-cost types relative to *Hard* and *Too Hard*. The exact comparison in Proposition 4 concerns two types whose cost functions differ by a proportional scaling. In particular, taking *Too Hard* as the high-cost benchmark, the proposition applies to a comparison with type $t \in \{\text{EASY}, \text{DOABLE}, \text{HARD}\}$ when

$$\kappa_{\text{TOO HARD}}(\pi_M) = \gamma_t \kappa_t(\pi_M) \qquad \text{for every } \pi_M \in \Pi,$$

with $\gamma_t \geq 1$.

We do not impose this proportionality restriction, or any cost ordering across types, in the structural estimation. Instead, we recover costs separately for each perception type and use the estimates to evaluate the implications of Proposition 4.

D(iv) IDENTIFICATION OF THE COST

First, we estimate the population frequencies $\rho^{\mathcal{D}}(\cdot)$ with which different rules are extracted. Then, for every decision problem $\mathcal{D}$ and every pair of rules $\pi_M, \pi'_{M'}$ that receive positive probability, the first-order conditions of equation (1) imply

$$v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'}) = \kappa(\pi_M)\rho^{\mathcal{D}}(\pi_M) - \kappa(\pi'_{M'})\rho^{\mathcal{D}}(\pi'_{M'}).$$

Because the cost of a rule is assumed to be invariant across decision problems, these restrictions can be stacked across decision problems. The costs of the rules appearing in the resulting system are identified when the system has full column rank. A rule that never receives positive probability cannot be identified through these restrictions. With a single decision problem and n active rules, the pairwise restrictions have rank at most $n-1$, so one normalization is required. Multiple decision problems can instead yield point identification.

To see why, consider two rules π_M and $\pi'_{M'}$ that receive positive probability in two decision problems, $\mathcal{D}$ and $\mathcal{D}'$. Then we have two equations:

$$\rho^{\mathcal{D}}(\pi_M)\kappa(\pi_M) - \rho^{\mathcal{D}}(\pi'_{M'})\kappa(\pi'_{M'}) = v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'}),$$
$$\rho^{\mathcal{D}'}(\pi_M)\kappa(\pi_M) - \rho^{\mathcal{D}'}(\pi'_{M'})\kappa(\pi'_{M'}) = v^{\mathcal{D}'}(\pi_M) - v^{\mathcal{D}'}(\pi'_{M'}),$$

and the two costs are identified if

$$\frac{\rho^{\mathcal{D}}(\pi_M)}{\rho^{\mathcal{D}}(\pi'_{M'})} \neq \frac{\rho^{\mathcal{D}'}(\pi_M)}{\rho^{\mathcal{D}'}(\pi'_{M'})}.$$

Thus, identification requires variation in the relative extraction probabilities of the two rules across decision problems. This approach allows for more nuanced predictions and can generate overidentifying restrictions when the stacked system contains more independent restrictions than cost parameters.

When the rule-specificity and rank conditions described above hold, the cost function can be estimated and additional theoretical predictions can be tested. In particular, given estimates of the cost function, the set of available rules Π, and the values of rules, the consideration set can be uniquely pinned down. This, in turn, allows us to (i) quantitatively evaluate the impact of changes in rule values and the comparative statics characterized in the theoretical propositions; and (ii) derive more nuanced predictions in settings that fall outside the sufficient conditions established in those propositions. First, to illustrate these points more formally, suppose that the consideration set remains unchanged, $\Pi^{\mathcal{D}} = \Pi^{\mathcal{D}'}$, and consider the expression for $\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M)$:

$$\begin{aligned}\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M) = \frac{1}{\kappa(\pi_M)} \Big[&\Big(\Delta^{\pi_M}(\mathcal{D},\mathcal{D}') - 1\Big) v^{\mathcal{D}}(\pi_M) \\ &- \sum_{\pi'_{M'} \in \Pi^{\mathcal{D}}} \Big(\Delta^{\pi'_{M'}}(\mathcal{D},\mathcal{D}') - 1\Big) v^{\mathcal{D}}(\pi'_{M'}) \frac{\frac{1}{\kappa(\pi'_{M'})}}{\kappa(\Pi^{\mathcal{D}})} \Big].\end{aligned} \tag{13}$$

Quantitative evaluation of equation (13) requires knowledge of the magnitude of the cost function, but allows us to extend predictions beyond the cases pinned down by the sufficient conditions in, e.g., Proposition 1. Second, note that the difference $\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M)$ depends both on how rule values change across decision problems and, potentially, on changes in the consideration set. For instance, if more rules enter the consideration set in $\mathcal{D}'$ than in $\mathcal{D}$, that is, if $\Pi^{\mathcal{D}'} \supset \Pi^{\mathcal{D}}$, then $\rho^{\mathcal{D}'}(\pi_M) - \rho^{\mathcal{D}}(\pi_M)$ may decrease due to a diffusion of choice probabilities, even when the value of rule π_M increases. By estimating the cost, we are able to characterize the consideration set ex ante, and the framework yields ex ante predictions about which force dominates.

Estimation. We implement the stacked identification procedure described above using the two- and three-light experiments. We treat each experiment–treatment–condition as a decision problem and estimate rule-extraction frequencies from the posterior rule assignments. For each perception type t, we stack the pairwise restrictions across decision problems and estimate a single type-specific cost vector by solving

$$\widehat{\kappa}_t \in \arg\min_{\kappa_t(\cdot)\geq 0} \sum_{\mathcal{D}} \sum_{\{\pi_M, \pi'_{M'}\} \subseteq \Pi_t^{\mathcal{D}}} \left[\rho_t^{\mathcal{D}}(\pi_M)\kappa_t(\pi_M) - \rho_t^{\mathcal{D}}(\pi'_{M'})\kappa_t(\pi'_{M'}) - \left(v^{\mathcal{D}}(\pi_M) - v^{\mathcal{D}}(\pi'_{M'})\right)\right]^2.$$

Thus, rule costs are invariant across decision problems within a perception type but may differ freely across types. To align structurally equivalent rules across experiments, we assign the red, blue, and green one-variable rules a common cost parameter. We likewise map the red–blue implementations of Inhibit, Either, Joint, and Or to their corresponding rule categories. Other experiment-specific rules enter the stacked system as nuisance cost parameters but are not reported. We require the stacked system to have full column rank and impose nonnegative costs. Confidence intervals are obtained from 2,000 subject-level Bayesian-bootstrap draws.

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